Integrated Enterprise Network Using Cisco Packet Tracer

CCNA Networking and Cybersecurity Project

Project Documentation

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Topology reference: CCNA_Project.pkt (Cisco Packet Tracer 9.0)

Certificate

This is to certify that the project report entitled **"Integrated Enterprise Network Using Cisco Packet Tracer"** is a bonafide work carried out by **Mohammad Umar Naeemullah Shaikh**, in partial fulfilment of the requirements of **Cyber Security & Ethical Hacking with AI** at **IT-Vedant (Dadar)** during the academic year **2026**.

Declaration

I, declare that this project report, "Integrated Enterprise Network Using Cisco Packet Tracer", is my own work. The network topology was built by me in Cisco Packet Tracer; the addressing plan, configuration and documentation were prepared for this project. Sources consulted are listed in the References. Test results are recorded only where they have actually been obtained.

Acknowledgement

I thank my project guide, **Kadir Ail** , and the faculty of the **IT-Vedant (Dadar)** for their guidance and support. I also thank the Cisco Networking Academy for the learning material and Cisco Packet Tracer, and my family and classmates for their encouragement.

Abstract

This project presents the design and documentation of an integrated enterprise network in Cisco Packet Tracer 9.0. The network connects a Main site, three Core routers, a Head Office, a Branch Office, a School and a College using 17 Cisco 2811 routers, 27 Cisco 2960 switches, 156 PCs, 28 laptops, 14 printers and 16 servers. Routers are linked by serial /30 subnets taken from 100.1.0.0/24, and each LAN is a /28 taken from the site range (10.1.0.0 - 10.4.0.0).

Routing uses multi-area OSPF, with areas aligned to the three Core routers, and eBGP between Core-R1 and the four Head Office routers. DHCP is served by the site routers, DNS (project.local), HTTP and HTTPS are served from the central server LAN 100.0.0.0/24, and Head Office uses NAT with PAT. Security controls include extended and standard ACLs, SSH version 2, secret passwords, VTY restrictions, banners, and Spanning Tree protections.

The document records discrepancies between the brief, the screenshot and the project file (for example 17 routers rather than 16, and an ATM site present only in the file). It supplies device-specific CLI for every router and switch, a verification and test plan with expected results, a troubleshooting guide and 54 viva questions. Actual test results are marked "To be verified" until the configuration is run.

Introduction

Modern organisations depend on an enterprise network that connects many sites, keeps traffic fast and reliable, and protects data from misuse. This project designs, documents and secures an **Integrated Enterprise Network** built in Cisco Packet Tracer 9.0. The network joins a central Main site, three Core routers, a Head Office, a Branch Office, a School and a College into one routed internetwork that carries data for hundreds of end devices.

The topology documented here is the topology that exists in the author's Packet Tracer project (CCNA_Project.pkt) and the screenshot of that project. No device was added, removed or re-cabled in this documentation. Where the brief, the screenshot and the project file disagree, the disagreement is recorded as a topology verification issue in Section 12.3 rather than silently corrected.

The design applies the CCNA 200-301 syllabus: IPv4 subnetting with VLSM, OSPF multi-area routing, eBGP, DHCP, DNS, HTTP/HTTPS, NAT/PAT, ACLs, SSH, and Spanning Tree Protocol, together with device hardening (secret passwords, banners, VTY security, port-level controls).

Problem Statement

A multi-site organisation needs one network where Head Office, Branch Office, School and College users can reach shared central services (DNS, web, backup), receive addresses automatically, and communicate across sites without manual configuration of every host. At the same time, sensitive networks must be protected from untrusted ones, administrators must manage devices only over encrypted channels, and the design must scale without re-addressing.

The supplied Packet Tracer project contains the physical cabling but no configuration: every router interface is administratively down, devices still carry the default hostname, no IP addresses exist on any host, and the DNS and DHCP services on the servers are switched off. The problem is therefore to produce a complete, internally consistent addressing, routing, services and security design for the existing cabling, and to document it so that it can be applied and verified device by device.

Objectives

- Document the actual topology (devices, links, interfaces) without altering it, and record every discrepancy.
- Create a VLSM addressing plan using the requested ranges (100.0.0.0, 100.1.0.0, 10.1.0.0 - 10.4.0.0) with /30 point-to-point serial subnets and /28 LAN subnets.
- Implement OSPF (multi-area) across all 17 routers and eBGP between Core-R1 and the Head Office routers.
- Provide DHCP, DNS, HTTP and HTTPS services to all sites from the central server LAN.
- Apply NAT/PAT at Head Office and ACL policy at Head Office, Branch Office and College.
- Secure all routers and switches with SSH v2, secret passwords, banners, VTY restrictions and STP protections.
- Provide device-specific Cisco IOS commands, verification commands, a test plan and a viva question bank.

Scope

In scope: the 17-router baseline shown in the screenshot (1 Main, 3 Core, 4 Head Office, 3 Branch Office, 3 School, 3 College routers), their 27 switches and all end devices; routing, services, security and verification for that baseline.

Out of scope / not documented as part of the baseline: the ATM-R1, ATM-R2, Switch18 - Switch21, Server8 and Server17 devices that appear only in the .pkt file (see V2 in Section 12.3); wireless, IPv6, VoIP and VLAN segmentation (no VLANs other than the default VLAN 1 exist in the project); real Internet connectivity.

Hardware / Software Requirements

Item	Specification (from the project file)	Use
Simulator	Cisco Packet Tracer 9.0 (project saved with version 9.0.0.0810)	Network simulation
Routers	Cisco 2811 (IOS 15.1) with Serial1/0 - 1/7, FastEthernet0/0 and 0/1	Routing, NAT, DHCP, ACL, BGP
Switches	Cisco 2960-24TT (24 x FastEthernet + 2 x GigabitEthernet)	Layer 2 access
End devices	PC-PT, Laptop-PT, Printer-PT, Server-PT	Clients, printers, services
Cables	Serial DCE/DTE (router-to-router); copper straight-through (router-to-switch and switch-to-host)	Connectivity
Host computer	Any PC able to run Packet Tracer 9.0 (64-bit OS, 4 GB RAM or more recommended)	Running the simulation
Documentation	Word processor; this document and PacketTracer_CLI_All_Devices.txt	Submission

Network Architecture

1 Hierarchical design

The network is a three-tier tree. The **Main router** is the root and holds the central server LAN. Three **Core routers** connect to Main over serial links. Each Core router feeds a group of **site routers**, and each site router feeds two switches over FastEthernet, which carry the end devices.

```
Main - R  (100.0.0.0/24 server LAN)
|-- S1/0 -- Core-R1 -- Head Office: HO-R1, HO-R2, HO-R3, HO-R4
|              \-- Branch Office: BO-R1
|-- S1/1 -- Core-R2 -- Branch Office: BO-R2, BO-R3
|              \-- School: SHC-R1, SHC-R2, SHC-R3
\-- S1/2 -- Core-R3 -- College: COL-R1, COL-R2, COL-R3

Every site router: Fa0/0 -> Switch A, Fa0/1 -> Switch B -> PCs / laptops / printers
```

The tree has no redundant router-to-router path and no switch-to-switch link. This is important for two reasons: every failure of a serial link isolates the sites below it, and Spanning Tree has no loop to block (Section 27).

2 Role of each layer

Tier	Routers	Role
Main	Main - R	Root of the tree; hosts DNS, web and backup servers on 100.0.0.0/24; OSPF Area 0
Core	Core-R1, Core-R2, Core-R3	Aggregate site routers; OSPF Area Border Routers (ABR); Core-R1 is the eBGP hub for Head Office
Distribution/site	HO-R1..4, BO-R1..3, SHC-R1..3, COL-R1..3	Default gateway, DHCP server, ACL/NAT enforcement for their LANs
Access	27 x 2960-24TT	Connect end devices; port-level protection

3 Topology verification issues

The brief, the screenshot and the project file were compared. The table lists every discrepancy found. Nothing was removed or invented to hide them.

ID	Item	Finding	Handling in this document
V1	Router count	Brief states 16 routers; the logical list adds up to 17 (1+3+4+3+3+3). The screenshot shows 17 routers. The .pkt file contains 19.	Baseline = 17. The "16" in the brief is a miscount.
V2	ATM site (.pkt only)	The .pkt contains ATM-R1, ATM-R2 (linked to Core-R3 S1/4 and S1/5), Switch18 - Switch21 (no end devices), plus Server8 and Server17 (uncabled). None appear in the screenshot or the brief.	Excluded from the baseline. Core-R3 S1/4 and S1/5 left unused. Can be added on request.
V3	Switch count	Brief: 27. Screenshot: 27 (Switch0 - Switch17, Switch22 - Switch30). .pkt: 31 (adds the four ATM switches).	Baseline = 27.
V4	Server count and cabling	Brief: 16. Screenshot: 16 named servers. .pkt: 18. In the .pkt only three servers (at Main) are cabled; the 13 per-router servers (HO-R1-Server, ...) have no cable.	16 servers documented. 13 are listed as NOT CABLED with a proposed address (Section 18.4).
V5	PC count	Brief: 154. Names in the screenshot and .pkt run HO-PC1..48, BO-PC1..36, SCH-PC1..42, COL-PC1..30 = 156.	156 PCs documented.
V6	COL-PC29	Present in screenshot beside Switch17; no cable in the .pkt. Switch17 has 4 hosts while the other College switches have 5.	Assumed on Switch17 Fa0/6. Requires verification in Cisco Packet Tracer.
V7	Naming	Router labels use "Main - R", "Core-R#", "SHC-R#"; School hosts use "SCH-"; laptops are "H0-Laptop" (zero) not "HO-".	Labels kept as in the topology. CLI hostnames use upper case without spaces (MAIN-R, CORE-R1, SHC-R1).
V8	Configuration state	All router interfaces are shut down, hostnames are "Router", no host has an IP, DNS and DHCP services are off.	All configuration here is a design to be applied.
V9	Server identity	Screenshot labels (e.g. HO-R1-Server) differ from .pkt names (Server10). Matching was done by canvas position. The three Main servers (Server0, Main-Server, Backup-Server) could not be mapped to .pkt names Server0 - Server2 with certainty.	Mapping shown in Section 18.4. Requires verification in Cisco Packet Tracer.
V10	Address ranges	Brief gives flat ranges (e.g. 10.1.0.2 - 10.1.0.254) per site, but each site has several isolated switch segments.	Each site /24 is split into /28 LANs (Section 15). Hosts still use the first usable address as the gateway.
V11	Printers / laptops	Brief: 14 printers, 28 laptops. Found: 14 and 28.	Match.

Complete Topology

Figure 1 is the author's own Packet Tracer screenshot, used as the primary topology reference. The colour regions are Head Office (green), College (cyan), Branch Office (orange) and School (yellow).

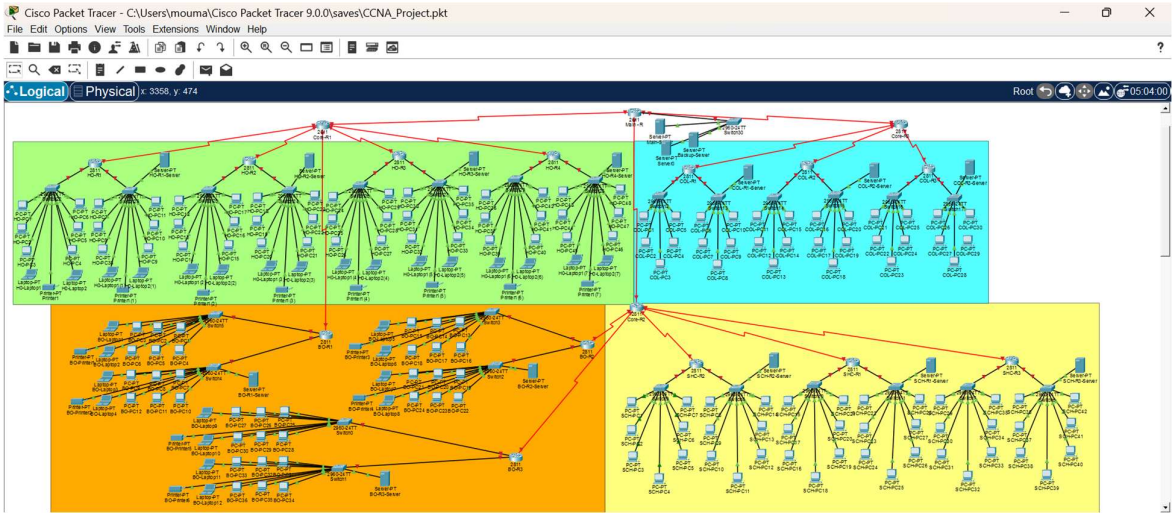


Figure 1: Integrated Enterprise Network - Packet Tracer logical view (author's screenshot)

1 Router-to-router links (serial)

All 16 serial links come from the project file. On every link the upstream (parent) router is the DCE end and carries the clock rate; the downstream router is DTE.

#	Parent (DCE)	Parent if	Child (DTE)	Child if
1	MAIN-R	S1/0	CORE-R1	S1/0
2	CORE-R1	S1/1	HO-R1	S1/0
3	CORE-R1	S1/2	HO-R2	S1/0
4	CORE-R1	S1/3	HO-R3	S1/0
5	CORE-R1	S1/4	HO-R4	S1/0
6	CORE-R1	S1/5	BO-R1	S1/0
7	MAIN-R	S1/1	CORE-R2	S1/0
8	MAIN-R	S1/2	CORE-R3	S1/0
9	CORE-R2	S1/1	BO-R2	S1/0
10	CORE-R2	S1/2	BO-R3	S1/0
11	CORE-R2	S1/3	SHC-R1	S1/0
12	CORE-R2	S1/4	SHC-R2	S1/0
13	CORE-R2	S1/5	SHC-R3	S1/0
14	CORE-R3	S1/1	COL-R1	S1/0
15	CORE-R3	S1/2	COL-R2	S1/0
16	CORE-R3	S1/3	COL-R3	S1/0

2 Router-to-switch links (FastEthernet)

Router	Fa0/0 -> switch	Fa0/1 -> switch	Site
MAIN-R	Switch30 (MAIN-SW1)	(unused)	Main
HO-R1	Switch22 (HO-R1-SWA)	Switch23 (HO-R1-SWB)	Head Office
HO-R2	Switch24 (HO-R2-SWA)	Switch25 (HO-R2-SWB)	Head Office
HO-R3	Switch26 (HO-R3-SWA)	Switch27 (HO-R3-SWB)	Head Office
HO-R4	Switch28 (HO-R4-SWA)	Switch29 (HO-R4-SWB)	Head Office

Router	Fa0/0 -> switch	Fa0/1 -> switch	Site
BO-R1	Switch4 (BO-R1-SWA)	Switch5 (BO-R1-SWB)	Branch Office
BO-R2	Switch2 (BO-R2-SWA)	Switch3 (BO-R2-SWB)	Branch Office
BO-R3	Switch0 (BO-R3-SWA)	Switch1 (BO-R3-SWB)	Branch Office
SHC-R1	Switch10 (SHC-R1-SWA)	Switch11 (SHC-R1-SWB)	School
SHC-R2	Switch8 (SHC-R2-SWA)	Switch9 (SHC-R2-SWB)	School
SHC-R3	Switch6 (SHC-R3-SWA)	Switch7 (SHC-R3-SWB)	School
COL-R1	Switch12 (COL-R1-SWA)	Switch13 (COL-R1-SWB)	College
COL-R2	Switch14 (COL-R2-SWA)	Switch15 (COL-R2-SWB)	College
COL-R3	Switch16 (COL-R3-SWA)	Switch17 (COL-R3-SWB)	College

Core routers have no LAN interfaces: their FastEthernet ports are unused. Main - R uses only Fa0/0 (to Switch30).

Device Inventory

1 Stated versus verified counts

Device	Verified
Routers	17
Switches	27
PCs	156
Servers	16
Printers	14
Laptops	28

2 Device inventory

Device Type	Quantity	Purpose	Location
Routers	17	Inter-site routing, gateways, DHCP, NAT, ACL, BGP	Main 1, Core 3, Head Office 4, Branch Office 3, School 3, College 3
Switches	27	Layer 2 access switching, STP	Main 1, Head Office 8, Branch Office 6, School 6, College 6
PCs	156	User workstations (DHCP clients)	Head Office 48, Branch Office 36, School 42, College 30
Servers	16	DNS, web (HTTP/HTTPS), backup; per-site servers	Main 3, Head Office 4, Branch Office 3, School 3, College 3
Printers	14	Network printers (static addresses)	Head Office 8, Branch Office 6
Laptops	28	Mobile clients (DHCP)	Head Office 16, Branch Office 12

3 Per-site matrix

Site	Routers	Switches	PCs	Laptops	Printers	Servers
Main	1	1	0	0	0	3
Core	3	0	0	0	0	0
Head Office	4	8	48	16	8	4
Branch Office	3	6	36	12	6	3
School	3	6	42	0	0	3
College	3	6	30	0	0	3
Total	17	27	156	28	14	16

IP Addressing Scheme

1 Design rules

- **Serial links use /30 subnets** carved sequentially from the Core range 100.1.0.0/24 (16 links x 4 addresses = 100.1.0.0 - 100.1.0.63). The first usable address belongs to the upstream (DCE) router, the second to the downstream router.
- **Main server LAN** is 100.0.0.0/24. 100.0.0.1 (the address requested for Main) is the Fa0/0 gateway of Main - R.
- **Site LANs** use the requested /24 of each site (10.1.0.0 Head Office, 10.2.0.0 Branch, 10.3.0.0 School, 10.4.0.0 College) divided into /28 subnets (14 usable hosts), one per switch segment, in router order (Fa0/0 then Fa0/1).
- In every /28: **.1 = router gateway, .2 = switch management (VLAN 1 SVI)**, end devices from .3 upward in switch-port order (Fa0/2 = .3, Fa0/3 = .4, ...).
- Unused space in each site /24 (e.g. 10.1.0.128 - 10.1.0.255) is reserved for growth.

2 Address blocks

Block	Prefix	Used for	Allocated
Main	100.0.0.0/24	Central server LAN	100.0.0.0/24
Core / WAN	100.1.0.0/24	16 serial /30 links	100.1.0.0 - 100.1.0.63
Head Office	10.1.0.0/24	8 LANs (/28)	10.1.0.0 - 10.1.0.127
Branch Office	10.2.0.0/24	6 LANs (/28)	10.2.0.0 - 10.2.0.95
School	10.3.0.0/24	6 LANs (/28)	10.3.0.0 - 10.3.0.95
College	10.4.0.0/24	6 LANs (/28)	10.4.0.0 - 10.4.0.95

3 Serial /30 subnets

#	Network	Mask	Upstream (DCE) IP	Downstream IP	Link
1	100.1.0.0/30	255.255.255.252	100.1.0.1	100.1.0.2	MAIN-R S1/0 - CORE-R1 S1/0
2	100.1.0.4/30	255.255.255.252	100.1.0.5	100.1.0.6	CORE-R1 S1/1 - HO-R1 S1/0
3	100.1.0.8/30	255.255.255.252	100.1.0.9	100.1.0.10	CORE-R1 S1/2 - HO-R2 S1/0
4	100.1.0.12/30	255.255.255.252	100.1.0.13	100.1.0.14	CORE-R1 S1/3 - HO-R3 S1/0
5	100.1.0.16/30	255.255.255.252	100.1.0.17	100.1.0.18	CORE-R1 S1/4 - HO-R4 S1/0
6	100.1.0.20/30	255.255.255.252	100.1.0.21	100.1.0.22	CORE-R1 S1/5 - BO-R1 S1/0
7	100.1.0.24/30	255.255.255.252	100.1.0.25	100.1.0.26	MAIN-R S1/1 - CORE-R2 S1/0
8	100.1.0.28/30	255.255.255.252	100.1.0.29	100.1.0.30	MAIN-R S1/2 - CORE-R3 S1/0
9	100.1.0.32/30	255.255.255.252	100.1.0.33	100.1.0.34	CORE-R2 S1/1 - BO-R2 S1/0
10	100.1.0.36/30	255.255.255.252	100.1.0.37	100.1.0.38	CORE-R2 S1/2 - BO-R3 S1/0
11	100.1.0.40/30	255.255.255.252	100.1.0.41	100.1.0.42	CORE-R2 S1/3 - SHC-R1 S1/0
12	100.1.0.44/30	255.255.255.252	100.1.0.45	100.1.0.46	CORE-R2 S1/4 - SHC-R2 S1/0
13	100.1.0.48/30	255.255.255.252	100.1.0.49	100.1.0.50	CORE-R2 S1/5 - SHC-R3 S1/0
14	100.1.0.52/30	255.255.255.252	100.1.0.53	100.1.0.54	CORE-R3 S1/1 - COL-R1 S1/0
15	100.1.0.56/30	255.255.255.252	100.1.0.57	100.1.0.58	CORE-R3 S1/2 - COL-R2 S1/0
16	100.1.0.60/30	255.255.255.252	100.1.0.61	100.1.0.62	CORE-R3 S1/3 - COL-R3 S1/0

4 LAN subnets

Router / interface	Switch (hostname)	Network	Mask	Gateway	Switch SVI	Usable range
MAIN-R F0/0	Switch30 (MAIN-SW1)	100.0.0.0/24	255.255.255.0	100.0.0.1	100.0.0.5	100.0.0.1 - 100.0.0.254
HO-R1 F0/0	Switch22 (HO-R1-SWA)	10.1.0.0/28	255.255.255.240	10.1.0.1	10.1.0.2	10.1.0.1 - 10.1.0.14
HO-R1 F0/1	Switch23 (HO-R1-SWB)	10.1.0.16/28	255.255.255.240	10.1.0.17	10.1.0.18	10.1.0.17 - 10.1.0.30
HO-R2 F0/0	Switch24 (HO-R2-SWA)	10.1.0.32/28	255.255.255.240	10.1.0.33	10.1.0.34	10.1.0.33 - 10.1.0.46
HO-R2 F0/1	Switch25 (HO-R2-SWB)	10.1.0.48/28	255.255.255.240	10.1.0.49	10.1.0.50	10.1.0.49 - 10.1.0.62

Router / interface	Switch (hostname)	Network	Mask	Gateway	Switch SVI	Usable range
HO-R3 F0/0	Switch26 (HO-R3-SWA)	10.1.0.64/28	255.255.255.240	10.1.0.65	10.1.0.66	10.1.0.65 - 10.1.0.78
HO-R3 F0/1	Switch27 (HO-R3-SWB)	10.1.0.80/28	255.255.255.240	10.1.0.81	10.1.0.82	10.1.0.81 - 10.1.0.94
HO-R4 F0/0	Switch28 (HO-R4-SWA)	10.1.0.96/28	255.255.255.240	10.1.0.97	10.1.0.98	10.1.0.97 - 10.1.0.110
HO-R4 F0/1	Switch29 (HO-R4-SWB)	10.1.0.112/28	255.255.255.240	10.1.0.113	10.1.0.114	10.1.0.113 - 10.1.0.126
BO-R1 F0/0	Switch4 (BO-R1-SWA)	10.2.0.0/28	255.255.255.240	10.2.0.1	10.2.0.2	10.2.0.1 - 10.2.0.14
BO-R1 F0/1	Switch5 (BO-R1-SWB)	10.2.0.16/28	255.255.255.240	10.2.0.17	10.2.0.18	10.2.0.17 - 10.2.0.30
BO-R2 F0/0	Switch2 (BO-R2-SWA)	10.2.0.32/28	255.255.255.240	10.2.0.33	10.2.0.34	10.2.0.33 - 10.2.0.46
BO-R2 F0/1	Switch3 (BO-R2-SWB)	10.2.0.48/28	255.255.255.240	10.2.0.49	10.2.0.50	10.2.0.49 - 10.2.0.62
BO-R3 F0/0	Switch0 (BO-R3-SWA)	10.2.0.64/28	255.255.255.240	10.2.0.65	10.2.0.66	10.2.0.65 - 10.2.0.78
BO-R3 F0/1	Switch1 (BO-R3-SWB)	10.2.0.80/28	255.255.255.240	10.2.0.81	10.2.0.82	10.2.0.81 - 10.2.0.94
SHC-R1 F0/0	Switch10 (SHC-R1-SWA)	10.3.0.0/28	255.255.255.240	10.3.0.1	10.3.0.2	10.3.0.1 - 10.3.0.14
SHC-R1 F0/1	Switch11 (SHC-R1-SWB)	10.3.0.16/28	255.255.255.240	10.3.0.17	10.3.0.18	10.3.0.17 - 10.3.0.30
SHC-R2 F0/0	Switch8 (SHC-R2-SWA)	10.3.0.32/28	255.255.255.240	10.3.0.33	10.3.0.34	10.3.0.33 - 10.3.0.46
SHC-R2 F0/1	Switch9 (SHC-R2-SWB)	10.3.0.48/28	255.255.255.240	10.3.0.49	10.3.0.50	10.3.0.49 - 10.3.0.62
SHC-R3 F0/0	Switch6 (SHC-R3-SWA)	10.3.0.64/28	255.255.255.240	10.3.0.65	10.3.0.66	10.3.0.65 - 10.3.0.78
SHC-R3 F0/1	Switch7 (SHC-R3-SWB)	10.3.0.80/28	255.255.255.240	10.3.0.81	10.3.0.82	10.3.0.81 - 10.3.0.94
COL-R1 F0/0	Switch12 (COL-R1-SWA)	10.4.0.0/28	255.255.255.240	10.4.0.1	10.4.0.2	10.4.0.1 - 10.4.0.14
COL-R1 F0/1	Switch13 (COL-R1-SWB)	10.4.0.16/28	255.255.255.240	10.4.0.17	10.4.0.18	10.4.0.17 - 10.4.0.30
COL-R2 F0/0	Switch14 (COL-R2-SWA)	10.4.0.32/28	255.255.255.240	10.4.0.33	10.4.0.34	10.4.0.33 - 10.4.0.46
COL-R2 F0/1	Switch15 (COL-R2-SWB)	10.4.0.48/28	255.255.255.240	10.4.0.49	10.4.0.50	10.4.0.49 - 10.4.0.62
COL-R3 F0/0	Switch16 (COL-R3-SWA)	10.4.0.64/28	255.255.255.240	10.4.0.65	10.4.0.66	10.4.0.65 - 10.4.0.78
COL-R3 F0/1	Switch17 (COL-R3-SWB)	10.4.0.80/28	255.255.255.240	10.4.0.81	10.4.0.82	10.4.0.81 - 10.4.0.94

5 Router IDs

Router	Router ID	Router	Router ID
MAIN-R	1.1.1.1	CORE-R1	2.2.2.1
CORE-R2	2.2.2.2	CORE-R3	2.2.2.3
HO-R1	3.3.3.1	HO-R2	3.3.3.2
HO-R3	3.3.3.3	HO-R4	3.3.3.4
BO-R1	4.4.4.1	BO-R2	4.4.4.2
BO-R3	4.4.4.3	SHC-R1	5.5.5.1
SHC-R2	5.5.5.2	SHC-R3	5.5.5.3
COL-R1	6.6.6.1	COL-R2	6.6.6.2
COL-R3	6.6.6.3		

Router IDs are set manually (they are not interface addresses) so that OSPF and BGP identities stay stable. The digit pattern identifies the tier: 1.1.1.x Main, 2.2.2.x Core, 3.3.3.x Head Office, 4.4.4.x Branch, 5.5.5.x School, 6.6.6.x College.

Router Interface Table

Every interface that is cabled in the baseline is listed. Unlisted interfaces (for example Main - R S1/3 - S1/7, Core S1/4 - S1/7, site-router S1/1 - S1/7) are not connected and are left shut down.

Router	Int.	Connection to	IP address	Subnet mask	Network	Purpose
MAIN-R	S1/0	CORE-R1 S1/0	100.1.0.1	255.255.255.252	100.1.0.0/30	WAN downlink to CORE-R1
MAIN-R	S1/1	CORE-R2 S1/0	100.1.0.25	255.255.255.252	100.1.0.24/30	WAN downlink to CORE-R2
MAIN-R	S1/2	CORE-R3 S1/0	100.1.0.29	255.255.255.252	100.1.0.28/30	WAN downlink to CORE-R3
MAIN-R	F0/0	Switch30 (MAIN-SW1) F0/1	100.0.0.1	255.255.255.0	100.0.0.0/24	Gateway for central server LAN (DNS/Web/Backup)
CORE-R1	S1/0	MAIN-R S1/0	100.1.0.2	255.255.255.252	100.1.0.0/30	WAN uplink to MAIN-R
CORE-R1	S1/1	HO-R1 S1/0	100.1.0.5	255.255.255.252	100.1.0.4/30	WAN downlink to HO-R1
CORE-R1	S1/2	HO-R2 S1/0	100.1.0.9	255.255.255.252	100.1.0.8/30	WAN downlink to HO-R2
CORE-R1	S1/3	HO-R3 S1/0	100.1.0.13	255.255.255.252	100.1.0.12/30	WAN downlink to HO-R3
CORE-R1	S1/4	HO-R4 S1/0	100.1.0.17	255.255.255.252	100.1.0.16/30	WAN downlink to HO-R4
CORE-R1	S1/5	BO-R1 S1/0	100.1.0.21	255.255.255.252	100.1.0.20/30	WAN downlink to BO-R1
CORE-R2	S1/0	MAIN-R S1/1	100.1.0.26	255.255.255.252	100.1.0.24/30	WAN uplink to MAIN-R
CORE-R2	S1/1	BO-R2 S1/0	100.1.0.33	255.255.255.252	100.1.0.32/30	WAN downlink to BO-R2
CORE-R2	S1/2	BO-R3 S1/0	100.1.0.37	255.255.255.252	100.1.0.36/30	WAN downlink to BO-R3
CORE-R2	S1/3	SHC-R1 S1/0	100.1.0.41	255.255.255.252	100.1.0.40/30	WAN downlink to SHC-R1
CORE-R2	S1/4	SHC-R2 S1/0	100.1.0.45	255.255.255.252	100.1.0.44/30	WAN downlink to SHC-R2
CORE-R2	S1/5	SHC-R3 S1/0	100.1.0.49	255.255.255.252	100.1.0.48/30	WAN downlink to SHC-R3
CORE-R3	S1/0	MAIN-R S1/2	100.1.0.30	255.255.255.252	100.1.0.28/30	WAN uplink to MAIN-R
CORE-R3	S1/1	COL-R1 S1/0	100.1.0.53	255.255.255.252	100.1.0.52/30	WAN downlink to COL-R1
CORE-R3	S1/2	COL-R2 S1/0	100.1.0.57	255.255.255.252	100.1.0.56/30	WAN downlink to COL-R2
CORE-R3	S1/3	COL-R3 S1/0	100.1.0.61	255.255.255.252	100.1.0.60/30	WAN downlink to COL-R3
HO-R1	S1/0	CORE-R1 S1/1	100.1.0.6	255.255.255.252	100.1.0.4/30	WAN uplink to CORE-R1
HO-R1	F0/0	Switch22 (HO-R1-SWA) F0/1	10.1.0.1	255.255.255.240	10.1.0.0/28	Gateway for HO-R1-SWA LAN (Head Office)
HO-R1	F0/1	Switch23 (HO-R1-SWB) F0/1	10.1.0.17	255.255.255.240	10.1.0.16/28	Gateway for HO-R1-SWB LAN (Head Office)
HO-R2	S1/0	CORE-R1 S1/2	100.1.0.10	255.255.255.252	100.1.0.8/30	WAN uplink to CORE-R1
HO-R2	F0/0	Switch24 (HO-R2-SWA) F0/1	10.1.0.33	255.255.255.240	10.1.0.32/28	Gateway for HO-R2-SWA LAN (Head Office)
HO-R2	F0/1	Switch25 (HO-R2-SWB) F0/1	10.1.0.49	255.255.255.240	10.1.0.48/28	Gateway for HO-R2-SWB LAN (Head Office)
HO-R3	S1/0	CORE-R1 S1/3	100.1.0.14	255.255.255.252	100.1.0.12/30	WAN uplink to CORE-R1
HO-R3	F0/0	Switch26 (HO-R3-SWA) F0/1	10.1.0.65	255.255.255.240	10.1.0.64/28	Gateway for HO-R3-SWA LAN (Head Office)
HO-R3	F0/1	Switch27 (HO-R3-SWB) F0/1	10.1.0.81	255.255.255.240	10.1.0.80/28	Gateway for HO-R3-SWB LAN (Head Office)
HO-R4	S1/0	CORE-R1 S1/4	100.1.0.18	255.255.255.252	100.1.0.16/30	WAN uplink to CORE-R1
HO-R4	F0/0	Switch28 (HO-R4-SWA) F0/1	10.1.0.97	255.255.255.240	10.1.0.96/28	Gateway for HO-R4-SWA LAN (Head Office)
HO-R4	F0/1	Switch29 (HO-R4-SWB) F0/1	10.1.0.113	255.255.255.240	10.1.0.112/28	Gateway for HO-R4-SWB LAN (Head Office)
BO-R1	S1/0	CORE-R1 S1/5	100.1.0.22	255.255.255.252	100.1.0.20/30	WAN uplink to CORE-R1
BO-R1	F0/0	Switch4 (BO-R1-SWA) F0/1	10.2.0.1	255.255.255.240	10.2.0.0/28	Gateway for BO-R1-SWA LAN (Branch Office)
BO-R1	F0/1	Switch5 (BO-R1-SWB) F0/1	10.2.0.17	255.255.255.240	10.2.0.16/28	Gateway for BO-R1-SWB LAN (Branch Office)
BO-R2	S1/0	CORE-R2 S1/1	100.1.0.34	255.255.255.252	100.1.0.32/30	WAN uplink to CORE-R2
BO-R2	F0/0	Switch2 (BO-R2-SWA) F0/1	10.2.0.33	255.255.255.240	10.2.0.32/28	Gateway for BO-R2-SWA LAN (Branch Office)
BO-R2	F0/1	Switch3 (BO-R2-SWB) F0/1	10.2.0.49	255.255.255.240	10.2.0.48/28	Gateway for BO-R2-SWB LAN (Branch Office)
BO-R3	S1/0	CORE-R2 S1/2	100.1.0.38	255.255.255.252	100.1.0.36/30	WAN uplink to CORE-R2
BO-R3	F0/0	Switch0 (BO-R3-SWA) F0/1	10.2.0.65	255.255.255.240	10.2.0.64/28	Gateway for BO-R3-SWA LAN (Branch Office)
BO-R3	F0/1	Switch1 (BO-R3-SWB) F0/1	10.2.0.81	255.255.255.240	10.2.0.80/28	Gateway for BO-R3-SWB LAN (Branch Office)
SHC-R1	S1/0	CORE-R2 S1/3	100.1.0.42	255.255.255.252	100.1.0.40/30	WAN uplink to CORE-R2

Router	Int.	Connection to	IP address	Subnet mask	Network	Purpose
SHC-R1	F0/0	Switch10 (SHC-R1-SWA) F0/1	10.3.0.1	255.255.255.240	10.3.0.0/28	Gateway for SHC-R1-SWA LAN (School)
SHC-R1	F0/1	Switch11 (SHC-R1-SWB) F0/1	10.3.0.17	255.255.255.240	10.3.0.16/28	Gateway for SHC-R1-SWB LAN (School)
SHC-R2	S1/0	CORE-R2 S1/4	100.1.0.46	255.255.255.252	100.1.0.44/30	WAN uplink to CORE-R2
SHC-R2	F0/0	Switch8 (SHC-R2-SWA) F0/1	10.3.0.33	255.255.255.240	10.3.0.32/28	Gateway for SHC-R2-SWA LAN (School)
SHC-R2	F0/1	Switch9 (SHC-R2-SWB) F0/1	10.3.0.49	255.255.255.240	10.3.0.48/28	Gateway for SHC-R2-SWB LAN (School)
SHC-R3	S1/0	CORE-R2 S1/5	100.1.0.50	255.255.255.252	100.1.0.48/30	WAN uplink to CORE-R2
SHC-R3	F0/0	Switch6 (SHC-R3-SWA) F0/1	10.3.0.65	255.255.255.240	10.3.0.64/28	Gateway for SHC-R3-SWA LAN (School)
SHC-R3	F0/1	Switch7 (SHC-R3-SWB) F0/1	10.3.0.81	255.255.255.240	10.3.0.80/28	Gateway for SHC-R3-SWB LAN (School)
COL-R1	S1/0	CORE-R3 S1/1	100.1.0.54	255.255.255.252	100.1.0.52/30	WAN uplink to CORE-R3
COL-R1	F0/0	Switch12 (COL-R1-SWA) F0/1	10.4.0.1	255.255.255.240	10.4.0.0/28	Gateway for COL-R1-SWA LAN (College)
COL-R1	F0/1	Switch13 (COL-R1-SWB) F0/1	10.4.0.17	255.255.255.240	10.4.0.16/28	Gateway for COL-R1-SWB LAN (College)
COL-R2	S1/0	CORE-R3 S1/2	100.1.0.58	255.255.255.252	100.1.0.56/30	WAN uplink to CORE-R3
COL-R2	F0/0	Switch14 (COL-R2-SWA) F0/1	10.4.0.33	255.255.255.240	10.4.0.32/28	Gateway for COL-R2-SWA LAN (College)
COL-R2	F0/1	Switch15 (COL-R2-SWB) F0/1	10.4.0.49	255.255.255.240	10.4.0.48/28	Gateway for COL-R2-SWB LAN (College)
COL-R3	S1/0	CORE-R3 S1/3	100.1.0.62	255.255.255.252	100.1.0.60/30	WAN uplink to CORE-R3
COL-R3	F0/0	Switch16 (COL-R3-SWA) F0/1	10.4.0.65	255.255.255.240	10.4.0.64/28	Gateway for COL-R3-SWA LAN (College)
COL-R3	F0/1	Switch17 (COL-R3-SWB) F0/1	10.4.0.81	255.255.255.240	10.4.0.80/28	Gateway for COL-R3-SWB LAN (College)

Switch Addressing

Each switch has one management address on VLAN 1 (the only VLAN that exists in the project) and a default gateway pointing to its router. The uplink port is always Fa0/1.

PT label	Hostname	Uplink to	Mgmt IP (VLAN 1)	Mask	Default gateway
Switch30	MAIN-SW1	MAIN-R F0/0	100.0.0.5	255.255.255.0	100.0.0.1
Switch22	HO-R1-SWA	HO-R1 F0/0	10.1.0.2	255.255.255.240	10.1.0.1
Switch23	HO-R1-SWB	HO-R1 F0/1	10.1.0.18	255.255.255.240	10.1.0.17
Switch24	HO-R2-SWA	HO-R2 F0/0	10.1.0.34	255.255.255.240	10.1.0.33
Switch25	HO-R2-SWB	HO-R2 F0/1	10.1.0.50	255.255.255.240	10.1.0.49
Switch26	HO-R3-SWA	HO-R3 F0/0	10.1.0.66	255.255.255.240	10.1.0.65
Switch27	HO-R3-SWB	HO-R3 F0/1	10.1.0.82	255.255.255.240	10.1.0.81
Switch28	HO-R4-SWA	HO-R4 F0/0	10.1.0.98	255.255.255.240	10.1.0.97
Switch29	HO-R4-SWB	HO-R4 F0/1	10.1.0.114	255.255.255.240	10.1.0.113
Switch4	BO-R1-SWA	BO-R1 F0/0	10.2.0.2	255.255.255.240	10.2.0.1
Switch5	BO-R1-SWB	BO-R1 F0/1	10.2.0.18	255.255.255.240	10.2.0.17
Switch2	BO-R2-SWA	BO-R2 F0/0	10.2.0.34	255.255.255.240	10.2.0.33
Switch3	BO-R2-SWB	BO-R2 F0/1	10.2.0.50	255.255.255.240	10.2.0.49
Switch0	BO-R3-SWA	BO-R3 F0/0	10.2.0.66	255.255.255.240	10.2.0.65
Switch1	BO-R3-SWB	BO-R3 F0/1	10.2.0.82	255.255.255.240	10.2.0.81
Switch10	SHC-R1-SWA	SHC-R1 F0/0	10.3.0.2	255.255.255.240	10.3.0.1
Switch11	SHC-R1-SWB	SHC-R1 F0/1	10.3.0.18	255.255.255.240	10.3.0.17
Switch8	SHC-R2-SWA	SHC-R2 F0/0	10.3.0.34	255.255.255.240	10.3.0.33
Switch9	SHC-R2-SWB	SHC-R2 F0/1	10.3.0.50	255.255.255.240	10.3.0.49
Switch6	SHC-R3-SWA	SHC-R3 F0/0	10.3.0.66	255.255.255.240	10.3.0.65
Switch7	SHC-R3-SWB	SHC-R3 F0/1	10.3.0.82	255.255.255.240	10.3.0.81
Switch12	COL-R1-SWA	COL-R1 F0/0	10.4.0.2	255.255.255.240	10.4.0.1
Switch13	COL-R1-SWB	COL-R1 F0/1	10.4.0.18	255.255.255.240	10.4.0.17
Switch14	COL-R2-SWA	COL-R2 F0/0	10.4.0.34	255.255.255.240	10.4.0.33
Switch15	COL-R2-SWB	COL-R2 F0/1	10.4.0.50	255.255.255.240	10.4.0.49
Switch16	COL-R3-SWA	COL-R3 F0/0	10.4.0.66	255.255.255.240	10.4.0.65
Switch17	COL-R3-SWB	COL-R3 F0/1	10.4.0.82	255.255.255.240	10.4.0.81

End Device Addressing

1 Allocation method

The project file contains no IP configuration for any host, so a systematic plan is used instead of invented values. The **planned address** of a device equals the LAN network address + switch port number + 1 (for example Fa0/2 -> .3). PCs and laptops are DHCP clients: the router leases addresses from the pool in request order, so the lease a device actually receives may differ from its planned address. Printers and servers use the planned address as a static address, and these addresses are excluded from the DHCP pools. Read actual leases with show ip dhcp binding.

2 PC address table (156)

Device	Switch	Port	Planned IP	Gateway	Subnet	Method
HO-PC1	HO-R1-SWA	F0/2	10.1.0.3	10.1.0.1	10.1.0.0/28	DHCP
HO-PC6	HO-R1-SWA	F0/3	10.1.0.4	10.1.0.1	10.1.0.0/28	DHCP
HO-PC2	HO-R1-SWA	F0/4	10.1.0.5	10.1.0.1	10.1.0.0/28	DHCP
HO-PC5	HO-R1-SWA	F0/5	10.1.0.6	10.1.0.1	10.1.0.0/28	DHCP
HO-PC3	HO-R1-SWA	F0/6	10.1.0.7	10.1.0.1	10.1.0.0/28	DHCP
HO-PC4	HO-R1-SWA	F0/7	10.1.0.8	10.1.0.1	10.1.0.0/28	DHCP
HO-PC7	HO-R1-SWB	F0/2	10.1.0.19	10.1.0.17	10.1.0.16/28	DHCP
HO-PC8	HO-R1-SWB	F0/3	10.1.0.20	10.1.0.17	10.1.0.16/28	DHCP
HO-PC38	HO-R1-SWB	F0/4	10.1.0.21	10.1.0.17	10.1.0.16/28	DHCP
HO-PC9	HO-R1-SWB	F0/8	10.1.0.25	10.1.0.17	10.1.0.16/28	DHCP
HO-PC10	HO-R1-SWB	F0/9	10.1.0.26	10.1.0.17	10.1.0.16/28	DHCP
HO-PC11	HO-R1-SWB	F0/10	10.1.0.27	10.1.0.17	10.1.0.16/28	DHCP
HO-PC18	HO-R2-SWA	F0/2	10.1.0.35	10.1.0.33	10.1.0.32/28	DHCP
HO-PC19	HO-R2-SWA	F0/3	10.1.0.36	10.1.0.33	10.1.0.32/28	DHCP
HO-PC20	HO-R2-SWA	F0/4	10.1.0.37	10.1.0.33	10.1.0.32/28	DHCP
HO-PC21	HO-R2-SWA	F0/8	10.1.0.41	10.1.0.33	10.1.0.32/28	DHCP
HO-PC22	HO-R2-SWA	F0/9	10.1.0.42	10.1.0.33	10.1.0.32/28	DHCP
HO-PC23	HO-R2-SWA	F0/10	10.1.0.43	10.1.0.33	10.1.0.32/28	DHCP
HO-PC12	HO-R2-SWB	F0/2	10.1.0.51	10.1.0.49	10.1.0.48/28	DHCP
HO-PC13	HO-R2-SWB	F0/3	10.1.0.52	10.1.0.49	10.1.0.48/28	DHCP
HO-PC14	HO-R2-SWB	F0/4	10.1.0.53	10.1.0.49	10.1.0.48/28	DHCP
HO-PC15	HO-R2-SWB	F0/8	10.1.0.57	10.1.0.49	10.1.0.48/28	DHCP
HO-PC16	HO-R2-SWB	F0/9	10.1.0.58	10.1.0.49	10.1.0.48/28	DHCP
HO-PC17	HO-R2-SWB	F0/10	10.1.0.59	10.1.0.49	10.1.0.48/28	DHCP
HO-PC24	HO-R3-SWA	F0/2	10.1.0.67	10.1.0.65	10.1.0.64/28	DHCP
HO-PC25	HO-R3-SWA	F0/3	10.1.0.68	10.1.0.65	10.1.0.64/28	DHCP
HO-PC26	HO-R3-SWA	F0/4	10.1.0.69	10.1.0.65	10.1.0.64/28	DHCP
HO-PC27	HO-R3-SWA	F0/8	10.1.0.73	10.1.0.65	10.1.0.64/28	DHCP
HO-PC28	HO-R3-SWA	F0/9	10.1.0.74	10.1.0.65	10.1.0.64/28	DHCP
HO-PC29	HO-R3-SWA	F0/10	10.1.0.75	10.1.0.65	10.1.0.64/28	DHCP
HO-PC30	HO-R3-SWB	F0/2	10.1.0.83	10.1.0.81	10.1.0.80/28	DHCP
HO-PC31	HO-R3-SWB	F0/3	10.1.0.84	10.1.0.81	10.1.0.80/28	DHCP
HO-PC32	HO-R3-SWB	F0/4	10.1.0.85	10.1.0.81	10.1.0.80/28	DHCP
HO-PC33	HO-R3-SWB	F0/8	10.1.0.89	10.1.0.81	10.1.0.80/28	DHCP
HO-PC34	HO-R3-SWB	F0/9	10.1.0.90	10.1.0.81	10.1.0.80/28	DHCP
HO-PC35	HO-R3-SWB	F0/10	10.1.0.91	10.1.0.81	10.1.0.80/28	DHCP
HO-PC43	HO-R4-SWA	F0/2	10.1.0.99	10.1.0.97	10.1.0.96/28	DHCP
HO-PC44	HO-R4-SWA	F0/3	10.1.0.100	10.1.0.97	10.1.0.96/28	DHCP
HO-PC45	HO-R4-SWA	F0/4	10.1.0.101	10.1.0.97	10.1.0.96/28	DHCP
HO-PC46	HO-R4-SWA	F0/8	10.1.0.105	10.1.0.97	10.1.0.96/28	DHCP
HO-PC47	HO-R4-SWA	F0/9	10.1.0.106	10.1.0.97	10.1.0.96/28	DHCP
HO-PC48	HO-R4-SWA	F0/10	10.1.0.107	10.1.0.97	10.1.0.96/28	DHCP
HO-PC36	HO-R4-SWB	F0/2	10.1.0.115	10.1.0.113	10.1.0.112/28	DHCP
HO-PC37	HO-R4-SWB	F0/3	10.1.0.116	10.1.0.113	10.1.0.112/28	DHCP

Device	Switch	Port	Planned IP	Gateway	Subnet	Method
HO-PC39	HO-R4-SWB	F0/4	10.1.0.117	10.1.0.113	10.1.0.112/28	DHCP
HO-PC40	HO-R4-SWB	F0/8	10.1.0.121	10.1.0.113	10.1.0.112/28	DHCP
HO-PC41	HO-R4-SWB	F0/9	10.1.0.122	10.1.0.113	10.1.0.112/28	DHCP
HO-PC42	HO-R4-SWB	F0/10	10.1.0.123	10.1.0.113	10.1.0.112/28	DHCP
BO-PC7	BO-R1-SWA	F0/2	10.2.0.3	10.2.0.1	10.2.0.0/28	DHCP
BO-PC8	BO-R1-SWA	F0/3	10.2.0.4	10.2.0.1	10.2.0.0/28	DHCP
BO-PC9	BO-R1-SWA	F0/4	10.2.0.5	10.2.0.1	10.2.0.0/28	DHCP
BO-PC10	BO-R1-SWA	F0/5	10.2.0.6	10.2.0.1	10.2.0.0/28	DHCP
BO-PC11	BO-R1-SWA	F0/6	10.2.0.7	10.2.0.1	10.2.0.0/28	DHCP
BO-PC12	BO-R1-SWA	F0/7	10.2.0.8	10.2.0.1	10.2.0.0/28	DHCP
BO-PC1	BO-R1-SWB	F0/2	10.2.0.19	10.2.0.17	10.2.0.16/28	DHCP
BO-PC2	BO-R1-SWB	F0/3	10.2.0.20	10.2.0.17	10.2.0.16/28	DHCP
BO-PC4	BO-R1-SWB	F0/4	10.2.0.21	10.2.0.17	10.2.0.16/28	DHCP
BO-PC5	BO-R1-SWB	F0/5	10.2.0.22	10.2.0.17	10.2.0.16/28	DHCP
BO-PC3	BO-R1-SWB	F0/6	10.2.0.23	10.2.0.17	10.2.0.16/28	DHCP
BO-PC6	BO-R1-SWB	F0/7	10.2.0.24	10.2.0.17	10.2.0.16/28	DHCP
BO-PC19	BO-R2-SWA	F0/2	10.2.0.35	10.2.0.33	10.2.0.32/28	DHCP
BO-PC20	BO-R2-SWA	F0/3	10.2.0.36	10.2.0.33	10.2.0.32/28	DHCP
BO-PC21	BO-R2-SWA	F0/4	10.2.0.37	10.2.0.33	10.2.0.32/28	DHCP
BO-PC22	BO-R2-SWA	F0/6	10.2.0.39	10.2.0.33	10.2.0.32/28	DHCP
BO-PC23	BO-R2-SWA	F0/7	10.2.0.40	10.2.0.33	10.2.0.32/28	DHCP
BO-PC24	BO-R2-SWA	F0/8	10.2.0.41	10.2.0.33	10.2.0.32/28	DHCP
BO-PC13	BO-R2-SWB	F0/2	10.2.0.51	10.2.0.49	10.2.0.48/28	DHCP
BO-PC14	BO-R2-SWB	F0/3	10.2.0.52	10.2.0.49	10.2.0.48/28	DHCP
BO-PC15	BO-R2-SWB	F0/4	10.2.0.53	10.2.0.49	10.2.0.48/28	DHCP
BO-PC16	BO-R2-SWB	F0/5	10.2.0.54	10.2.0.49	10.2.0.48/28	DHCP
BO-PC17	BO-R2-SWB	F0/6	10.2.0.55	10.2.0.49	10.2.0.48/28	DHCP
BO-PC18	BO-R2-SWB	F0/7	10.2.0.56	10.2.0.49	10.2.0.48/28	DHCP
BO-PC25	BO-R3-SWA	F0/2	10.2.0.67	10.2.0.65	10.2.0.64/28	DHCP
BO-PC26	BO-R3-SWA	F0/3	10.2.0.68	10.2.0.65	10.2.0.64/28	DHCP
BO-PC27	BO-R3-SWA	F0/4	10.2.0.69	10.2.0.65	10.2.0.64/28	DHCP
BO-PC28	BO-R3-SWA	F0/5	10.2.0.70	10.2.0.65	10.2.0.64/28	DHCP
BO-PC29	BO-R3-SWA	F0/6	10.2.0.71	10.2.0.65	10.2.0.64/28	DHCP
BO-PC30	BO-R3-SWA	F0/7	10.2.0.72	10.2.0.65	10.2.0.64/28	DHCP
BO-PC31	BO-R3-SWB	F0/2	10.2.0.83	10.2.0.81	10.2.0.80/28	DHCP
BO-PC32	BO-R3-SWB	F0/3	10.2.0.84	10.2.0.81	10.2.0.80/28	DHCP
BO-PC33	BO-R3-SWB	F0/4	10.2.0.85	10.2.0.81	10.2.0.80/28	DHCP
BO-PC34	BO-R3-SWB	F0/5	10.2.0.86	10.2.0.81	10.2.0.80/28	DHCP
BO-PC35	BO-R3-SWB	F0/6	10.2.0.87	10.2.0.81	10.2.0.80/28	DHCP
BO-PC36	BO-R3-SWB	F0/7	10.2.0.88	10.2.0.81	10.2.0.80/28	DHCP
SCH-PC15	SHC-R1-SWA	F0/2	10.3.0.3	10.3.0.1	10.3.0.0/28	DHCP
SCH-PC17	SHC-R1-SWA	F0/3	10.3.0.4	10.3.0.1	10.3.0.0/28	DHCP
SCH-PC16	SHC-R1-SWA	F0/4	10.3.0.5	10.3.0.1	10.3.0.0/28	DHCP
SCH-PC18	SHC-R1-SWA	F0/5	10.3.0.6	10.3.0.1	10.3.0.0/28	DHCP
SCH-PC19	SHC-R1-SWA	F0/6	10.3.0.7	10.3.0.1	10.3.0.0/28	DHCP
SCH-PC20	SHC-R1-SWA	F0/7	10.3.0.8	10.3.0.1	10.3.0.0/28	DHCP
SCH-PC21	SHC-R1-SWA	F0/8	10.3.0.9	10.3.0.1	10.3.0.0/28	DHCP
SCH-PC22	SHC-R1-SWB	F0/2	10.3.0.19	10.3.0.17	10.3.0.16/28	DHCP
SCH-PC23	SHC-R1-SWB	F0/3	10.3.0.20	10.3.0.17	10.3.0.16/28	DHCP
SCH-PC24	SHC-R1-SWB	F0/4	10.3.0.21	10.3.0.17	10.3.0.16/28	DHCP
SCH-PC25	SHC-R1-SWB	F0/5	10.3.0.22	10.3.0.17	10.3.0.16/28	DHCP
SCH-PC26	SHC-R1-SWB	F0/6	10.3.0.23	10.3.0.17	10.3.0.16/28	DHCP
SCH-PC27	SHC-R1-SWB	F0/7	10.3.0.24	10.3.0.17	10.3.0.16/28	DHCP
SCH-PC28	SHC-R1-SWB	F0/8	10.3.0.25	10.3.0.17	10.3.0.16/28	DHCP
SCH-PC8	SHC-R2-SWA	F0/2	10.3.0.35	10.3.0.33	10.3.0.32/28	DHCP
SCH-PC9	SHC-R2-SWA	F0/3	10.3.0.36	10.3.0.33	10.3.0.32/28	DHCP

Device	Switch	Port	Planned IP	Gateway	Subnet	Method
SCH-PC10	SHC-R2-SWA	F0/4	10.3.0.37	10.3.0.33	10.3.0.32/28	DHCP
SCH-PC11	SHC-R2-SWA	F0/5	10.3.0.38	10.3.0.33	10.3.0.32/28	DHCP
SCH-PC12	SHC-R2-SWA	F0/6	10.3.0.39	10.3.0.33	10.3.0.32/28	DHCP
SCH-PC13	SHC-R2-SWA	F0/7	10.3.0.40	10.3.0.33	10.3.0.32/28	DHCP
SCH-PC14	SHC-R2-SWA	F0/8	10.3.0.41	10.3.0.33	10.3.0.32/28	DHCP
SCH-PC1	SHC-R2-SWB	F0/2	10.3.0.51	10.3.0.49	10.3.0.48/28	DHCP
SCH-PC2	SHC-R2-SWB	F0/3	10.3.0.52	10.3.0.49	10.3.0.48/28	DHCP
SCH-PC3	SHC-R2-SWB	F0/4	10.3.0.53	10.3.0.49	10.3.0.48/28	DHCP
SCH-PC4	SHC-R2-SWB	F0/5	10.3.0.54	10.3.0.49	10.3.0.48/28	DHCP
SCH-PC5	SHC-R2-SWB	F0/6	10.3.0.55	10.3.0.49	10.3.0.48/28	DHCP
SCH-PC6	SHC-R2-SWB	F0/7	10.3.0.56	10.3.0.49	10.3.0.48/28	DHCP
SCH-PC7	SHC-R2-SWB	F0/8	10.3.0.57	10.3.0.49	10.3.0.48/28	DHCP
SCH-PC36	SHC-R3-SWA	F0/2	10.3.0.67	10.3.0.65	10.3.0.64/28	DHCP
SCH-PC37	SHC-R3-SWA	F0/3	10.3.0.68	10.3.0.65	10.3.0.64/28	DHCP
SCH-PC38	SHC-R3-SWA	F0/4	10.3.0.69	10.3.0.65	10.3.0.64/28	DHCP
SCH-PC39	SHC-R3-SWA	F0/5	10.3.0.70	10.3.0.65	10.3.0.64/28	DHCP
SCH-PC40	SHC-R3-SWA	F0/6	10.3.0.71	10.3.0.65	10.3.0.64/28	DHCP
SCH-PC41	SHC-R3-SWA	F0/7	10.3.0.72	10.3.0.65	10.3.0.64/28	DHCP
SCH-PC42	SHC-R3-SWA	F0/8	10.3.0.73	10.3.0.65	10.3.0.64/28	DHCP
SCH-PC29	SHC-R3-SWB	F0/2	10.3.0.83	10.3.0.81	10.3.0.80/28	DHCP
SCH-PC30	SHC-R3-SWB	F0/3	10.3.0.84	10.3.0.81	10.3.0.80/28	DHCP
SCH-PC31	SHC-R3-SWB	F0/4	10.3.0.85	10.3.0.81	10.3.0.80/28	DHCP
SCH-PC32	SHC-R3-SWB	F0/5	10.3.0.86	10.3.0.81	10.3.0.80/28	DHCP
SCH-PC33	SHC-R3-SWB	F0/6	10.3.0.87	10.3.0.81	10.3.0.80/28	DHCP
SCH-PC34	SHC-R3-SWB	F0/7	10.3.0.88	10.3.0.81	10.3.0.80/28	DHCP
SCH-PC35	SHC-R3-SWB	F0/8	10.3.0.89	10.3.0.81	10.3.0.80/28	DHCP
COL-PC1	COL-R1-SWA	F0/2	10.4.0.3	10.4.0.1	10.4.0.0/28	DHCP
COL-PC2	COL-R1-SWA	F0/3	10.4.0.4	10.4.0.1	10.4.0.0/28	DHCP
COL-PC3	COL-R1-SWA	F0/4	10.4.0.5	10.4.0.1	10.4.0.0/28	DHCP
COL-PC4	COL-R1-SWA	F0/5	10.4.0.6	10.4.0.1	10.4.0.0/28	DHCP
COL-PC5	COL-R1-SWA	F0/6	10.4.0.7	10.4.0.1	10.4.0.0/28	DHCP
COL-PC6	COL-R1-SWB	F0/2	10.4.0.19	10.4.0.17	10.4.0.16/28	DHCP
COL-PC7	COL-R1-SWB	F0/3	10.4.0.20	10.4.0.17	10.4.0.16/28	DHCP
COL-PC8	COL-R1-SWB	F0/4	10.4.0.21	10.4.0.17	10.4.0.16/28	DHCP
COL-PC9	COL-R1-SWB	F0/5	10.4.0.22	10.4.0.17	10.4.0.16/28	DHCP
COL-PC10	COL-R1-SWB	F0/6	10.4.0.23	10.4.0.17	10.4.0.16/28	DHCP
COL-PC11	COL-R2-SWA	F0/2	10.4.0.35	10.4.0.33	10.4.0.32/28	DHCP
COL-PC12	COL-R2-SWA	F0/3	10.4.0.36	10.4.0.33	10.4.0.32/28	DHCP
COL-PC13	COL-R2-SWA	F0/4	10.4.0.37	10.4.0.33	10.4.0.32/28	DHCP
COL-PC14	COL-R2-SWA	F0/5	10.4.0.38	10.4.0.33	10.4.0.32/28	DHCP
COL-PC15	COL-R2-SWA	F0/6	10.4.0.39	10.4.0.33	10.4.0.32/28	DHCP
COL-PC16	COL-R2-SWB	F0/2	10.4.0.51	10.4.0.49	10.4.0.48/28	DHCP
COL-PC17	COL-R2-SWB	F0/3	10.4.0.52	10.4.0.49	10.4.0.48/28	DHCP
COL-PC18	COL-R2-SWB	F0/4	10.4.0.53	10.4.0.49	10.4.0.48/28	DHCP
COL-PC20	COL-R2-SWB	F0/5	10.4.0.54	10.4.0.49	10.4.0.48/28	DHCP
COL-PC19	COL-R2-SWB	F0/6	10.4.0.55	10.4.0.49	10.4.0.48/28	DHCP
COL-PC21	COL-R3-SWA	F0/2	10.4.0.67	10.4.0.65	10.4.0.64/28	DHCP
COL-PC22	COL-R3-SWA	F0/3	10.4.0.68	10.4.0.65	10.4.0.64/28	DHCP
COL-PC23	COL-R3-SWA	F0/4	10.4.0.69	10.4.0.65	10.4.0.64/28	DHCP
COL-PC25	COL-R3-SWA	F0/5	10.4.0.70	10.4.0.65	10.4.0.64/28	DHCP
COL-PC24	COL-R3-SWA	F0/6	10.4.0.71	10.4.0.65	10.4.0.64/28	DHCP
COL-PC26	COL-R3-SWB	F0/2	10.4.0.83	10.4.0.81	10.4.0.80/28	DHCP
COL-PC27	COL-R3-SWB	F0/3	10.4.0.84	10.4.0.81	10.4.0.80/28	DHCP
COL-PC28	COL-R3-SWB	F0/4	10.4.0.85	10.4.0.81	10.4.0.80/28	DHCP
COL-PC30	COL-R3-SWB	F0/5	10.4.0.86	10.4.0.81	10.4.0.80/28	DHCP
COL-PC29	COL-R3-SWB	F0/6	10.4.0.87	10.4.0.81	10.4.0.80/28	DHCP

Laptop address table (28)

Device	Switch	Port	Planned IP	Gateway	Subnet	Method
H0-Laptop1	HO-R1-SWA	F0/8	10.1.0.9	10.1.0.1	10.1.0.0/28	DHCP
H0-Laptop2	HO-R1-SWA	F0/9	10.1.0.10	10.1.0.1	10.1.0.0/28	DHCP
H0-Laptop1(1)	HO-R1-SWB	F0/5	10.1.0.22	10.1.0.17	10.1.0.16/28	DHCP
H0-Laptop2(1)	HO-R1-SWB	F0/7	10.1.0.24	10.1.0.17	10.1.0.16/28	DHCP
H0-Laptop1(3)	HO-R2-SWA	F0/5	10.1.0.38	10.1.0.33	10.1.0.32/28	DHCP
H0-Laptop2(3)	HO-R2-SWA	F0/7	10.1.0.40	10.1.0.33	10.1.0.32/28	DHCP
H0-Laptop1(2)	HO-R2-SWB	F0/5	10.1.0.54	10.1.0.49	10.1.0.48/28	DHCP
H0-Laptop2(2)	HO-R2-SWB	F0/7	10.1.0.56	10.1.0.49	10.1.0.48/28	DHCP
H0-Laptop1(4)	HO-R3-SWA	F0/5	10.1.0.70	10.1.0.65	10.1.0.64/28	DHCP
H0-Laptop2(4)	HO-R3-SWA	F0/7	10.1.0.72	10.1.0.65	10.1.0.64/28	DHCP
H0-Laptop1(5)	HO-R3-SWB	F0/5	10.1.0.86	10.1.0.81	10.1.0.80/28	DHCP
H0-Laptop2(5)	HO-R3-SWB	F0/7	10.1.0.88	10.1.0.81	10.1.0.80/28	DHCP
H0-Laptop1(7)	HO-R4-SWA	F0/5	10.1.0.102	10.1.0.97	10.1.0.96/28	DHCP
H0-Laptop2(7)	HO-R4-SWA	F0/7	10.1.0.104	10.1.0.97	10.1.0.96/28	DHCP
H0-Laptop1(6)	HO-R4-SWB	F0/5	10.1.0.118	10.1.0.113	10.1.0.112/28	DHCP
H0-Laptop2(6)	HO-R4-SWB	F0/7	10.1.0.120	10.1.0.113	10.1.0.112/28	DHCP
BO-Laptop3	BO-R1-SWA	F0/8	10.2.0.9	10.2.0.1	10.2.0.0/28	DHCP
BO-Laptop4	BO-R1-SWA	F0/9	10.2.0.10	10.2.0.1	10.2.0.0/28	DHCP
BO-Laptop1	BO-R1-SWB	F0/8	10.2.0.25	10.2.0.17	10.2.0.16/28	DHCP
BO-Laptop2	BO-R1-SWB	F0/9	10.2.0.26	10.2.0.17	10.2.0.16/28	DHCP
BO-Laptop7	BO-R2-SWA	F0/5	10.2.0.38	10.2.0.33	10.2.0.32/28	DHCP
BO-Laptop8	BO-R2-SWA	F0/9	10.2.0.42	10.2.0.33	10.2.0.32/28	DHCP
BO-Laptop6	BO-R2-SWB	F0/8	10.2.0.57	10.2.0.49	10.2.0.48/28	DHCP
BO-Laptop5	BO-R2-SWB	F0/9	10.2.0.58	10.2.0.49	10.2.0.48/28	DHCP
BO-Laptop9	BO-R3-SWA	F0/8	10.2.0.73	10.2.0.65	10.2.0.64/28	DHCP
BO-Laptop10	BO-R3-SWA	F0/9	10.2.0.74	10.2.0.65	10.2.0.64/28	DHCP
BO-Laptop11	BO-R3-SWB	F0/8	10.2.0.89	10.2.0.81	10.2.0.80/28	DHCP
BO-Laptop12	BO-R3-SWB	F0/9	10.2.0.90	10.2.0.81	10.2.0.80/28	DHCP

4 Printer address table (14)

Device	Switch	Port	Planned IP	Gateway	Subnet	Method
Printer1	HO-R1-SWA	F0/10	10.1.0.11	10.1.0.1	10.1.0.0/28	DHCP
Printer1(1)	HO-R1-SWB	F0/6	10.1.0.23	10.1.0.17	10.1.0.16/28	DHCP
Printer1(3)	HO-R2-SWA	F0/6	10.1.0.39	10.1.0.33	10.1.0.32/28	DHCP
Printer1(2)	HO-R2-SWB	F0/6	10.1.0.55	10.1.0.49	10.1.0.48/28	DHCP
Printer1(4)	HO-R3-SWA	F0/6	10.1.0.71	10.1.0.65	10.1.0.64/28	DHCP
Printer1(5)	HO-R3-SWB	F0/6	10.1.0.87	10.1.0.81	10.1.0.80/28	DHCP
Printer1(7)	HO-R4-SWA	F0/6	10.1.0.103	10.1.0.97	10.1.0.96/28	DHCP
Printer1(6)	HO-R4-SWB	F0/6	10.1.0.119	10.1.0.113	10.1.0.112/28	DHCP
BO-Printer2	BO-R1-SWA	F0/10	10.2.0.11	10.2.0.1	10.2.0.0/28	DHCP
BO-Printer1	BO-R1-SWB	F0/10	10.2.0.27	10.2.0.17	10.2.0.16/28	DHCP
BO-Printer4	BO-R2-SWA	F0/10	10.2.0.43	10.2.0.33	10.2.0.32/28	DHCP
BO-Printer3	BO-R2-SWB	F0/10	10.2.0.59	10.2.0.49	10.2.0.48/28	DHCP
BO-Printer5	BO-R3-SWA	F0/10	10.2.0.75	10.2.0.65	10.2.0.64/28	DHCP
BO-Printer6	BO-R3-SWB	F0/10	10.2.0.91	10.2.0.81	10.2.0.80/28	DHCP

5 Server address table (16)

Server (screenshot label)	.pkt name	Switch / port	IP address	Mask	Gateway	Status
Server0	Server0	MAIN-SW1 F0/2	100.0.0.2	255.255.255.0	100.0.0.1	Cabled
Main-Server	Server1	MAIN-SW1 F0/3	100.0.0.3	255.255.255.0	100.0.0.1	Cabled
Backup-Server	Server2	MAIN-SW1 F0/4	100.0.0.4	255.255.255.0	100.0.0.1	Cabled
HO-R1-Server	Server10	HO-R1-SWA F0/11	10.1.0.12	255.255.255.240	10.1.0.1	NOT CABLED - proposed
HO-R2-Server	Server11	HO-R2-SWA F0/11	10.1.0.44	255.255.255.240	10.1.0.33	NOT CABLED - proposed
HO-R3-Server	Server9	HO-R3-SWA F0/11	10.1.0.76	255.255.255.240	10.1.0.65	NOT CABLED - proposed
HO-R4-Server	Server7	HO-R4-SWA F0/11	10.1.0.108	255.255.255.240	10.1.0.97	NOT CABLED - proposed
BO-R1-Server	Server3	BO-R1-SWA F0/11	10.2.0.12	255.255.255.240	10.2.0.1	NOT CABLED - proposed
BO-R2-Server	Server4	BO-R2-SWA F0/11	10.2.0.44	255.255.255.240	10.2.0.33	NOT CABLED - proposed
BO-R3-Server	Server5	BO-R3-SWA F0/11	10.2.0.76	255.255.255.240	10.2.0.65	NOT CABLED - proposed
SHC-R1-Server	Server12	SHC-R1-SWA F0/9	10.3.0.10	255.255.255.240	10.3.0.1	NOT CABLED - proposed
SHC-R2-Server	Server6	SHC-R2-SWA F0/9	10.3.0.42	255.255.255.240	10.3.0.33	NOT CABLED - proposed
SHC-R3-Server	Server13	SHC-R3-SWA F0/9	10.3.0.74	255.255.255.240	10.3.0.65	NOT CABLED - proposed
COL-R1-Server	Server14	COL-R1-SWA F0/7	10.4.0.8	255.255.255.240	10.4.0.1	NOT CABLED - proposed
COL-R2-Server	Server15	COL-R2-SWA F0/7	10.4.0.40	255.255.255.240	10.4.0.33	NOT CABLED - proposed
COL-R3-Server	Server16	COL-R3-SWA F0/7	10.4.0.72	255.255.255.240	10.4.0.65	NOT CABLED - proposed

Server Configuration

Three servers are cabled, all on the Main server LAN behind Switch30. They provide the shared services for every site. Server-PT services are configured on the Services tab and the address on Desktop > IP Configuration.

Server	Role	IP / mask	Gateway	DNS	Services to enable
Server0	DNS server	100.0.0.2 / 255.255.255.0	100.0.0.1	100.0.0.2	DNS = On (records in Section 28); HTTP/HTTPS may stay on
Main-Server	Web server	100.0.0.3 / 255.255.255.0	100.0.0.1	100.0.0.2	HTTP = On, HTTPS = On (edit index.html); DNS = Off
Backup-Server	Backup (FTP/TFTP)	100.0.0.4 / 255.255.255.0	100.0.0.1	100.0.0.2	FTP = On, TFTP = On (store router configurations)

The 13 per-router servers (HO-R1-Server ... COL-R3-Server) are not cabled, so no service is required for the baseline. If they are cabled later, configure them with the address in Section 18.5 and the DNS server 100.0.0.2.

OSPF

1 Explanation

Open Shortest Path First (OSPF) is a link-state interior gateway protocol (RFC 2328). Each router floods link-state advertisements, builds an identical map of its area and runs Dijkstra's shortest-path-first algorithm to fill the routing table. OSPF converges quickly, supports VLSM, and scales through areas. Its administrative distance is 110. On serial (HDLC) links OSPF uses the point-to-point network type, so no designated router (DR/BDR) election takes place.

2 Design

Item	Value
Process ID	1 on every router (locally significant)
Area 0 (backbone)	Main LAN 100.0.0.0/24 and the three Main - Core serial links
Area 1	Core-R1 domain: Core-R1 to HO-R1..4 and BO-R1 links and all their LANs
Area 2	Core-R2 domain: Core-R2 to BO-R2, BO-R3, SHC-R1..3 links and their LANs
Area 3	Core-R3 domain: Core-R3 to COL-R1..3 links and their LANs
ABRs	Core-R1 (areas 0/1), Core-R2 (0/2), Core-R3 (0/3)
Router IDs	Manual: see Section 15.5
Interface cost	ip ospf cost 50 on every serial (2 Mbit/s clock rate: 100 Mbit/s reference / 2 Mbit/s); FastEthernet default cost 1
Passive interfaces	All LAN-facing FastEthernet interfaces (no hellos sent to hosts)

3 Expected neighbor relationships

#	Router A (ID)	Router B (ID)	Link network	Area	Expected state
1	MAIN-R (1.1.1.1)	CORE-R1 (2.2.2.1)	100.1.0.0/30	0	FULL/ -
2	CORE-R1 (2.2.2.1)	HO-R1 (3.3.3.1)	100.1.0.4/30	1	FULL/ -
3	CORE-R1 (2.2.2.1)	HO-R2 (3.3.3.2)	100.1.0.8/30	1	FULL/ -
4	CORE-R1 (2.2.2.1)	HO-R3 (3.3.3.3)	100.1.0.12/30	1	FULL/ -
5	CORE-R1 (2.2.2.1)	HO-R4 (3.3.3.4)	100.1.0.16/30	1	FULL/ -
6	CORE-R1 (2.2.2.1)	BO-R1 (4.4.4.1)	100.1.0.20/30	1	FULL/ -
7	MAIN-R (1.1.1.1)	CORE-R2 (2.2.2.2)	100.1.0.24/30	0	FULL/ -
8	MAIN-R (1.1.1.1)	CORE-R3 (2.2.2.3)	100.1.0.28/30	0	FULL/ -
9	CORE-R2 (2.2.2.2)	BO-R2 (4.4.4.2)	100.1.0.32/30	2	FULL/ -
10	CORE-R2 (2.2.2.2)	BO-R3 (4.4.4.3)	100.1.0.36/30	2	FULL/ -
11	CORE-R2 (2.2.2.2)	SHC-R1 (5.5.5.1)	100.1.0.40/30	2	FULL/ -
12	CORE-R2 (2.2.2.2)	SHC-R2 (5.5.5.2)	100.1.0.44/30	2	FULL/ -
13	CORE-R2 (2.2.2.2)	SHC-R3 (5.5.5.3)	100.1.0.48/30	2	FULL/ -
14	CORE-R3 (2.2.2.3)	COL-R1 (6.6.6.1)	100.1.0.52/30	3	FULL/ -
15	CORE-R3 (2.2.2.3)	COL-R2 (6.6.6.2)	100.1.0.56/30	3	FULL/ -
16	CORE-R3 (2.2.2.3)	COL-R3 (6.6.6.3)	100.1.0.60/30	3	FULL/ -

16 adjacencies are expected in total. For example, Core-R1 should show six neighbors (Main - R, HO-R1..4 and BO-R1) in state FULL, with "-" in the DR/BDR column because the links are point-to-point.

5 Verification commands

show ip ospf neighbor
show ip ospf

HO-R1

Physical Config CLI Attributes

```
HO-R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.2           0   FULL/  -        00:00:30    100.1.0.13      Serial1/0
HO-R1#show ip ospf
 Routing Process "ospf 1" with ID 1.1.1.5
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 It is an area border router
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 2. 2 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 1
        Area has no authentication
        SPF algorithm executed 3 times
        Area ranges are
        Number of LSA 43. Checksum Sum 0x16aecb
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
    Area 1
        Number of interfaces in this area is 2
        Area has no authentication
        SPF algorithm executed 3 times
        Area ranges are
        Number of LSA 42. Checksum Sum 0x134fe9
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

BO-R1

Physical Config CLI Attributes

```
BO-R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.2           0   FULL/  -        00:00:35    100.1.0.29      Serial1/0
BO-R1#show ip ospf
 Routing Process "ospf 1" with ID 1.1.1.9
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 It is an area border router
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 2. 2 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 1
        Area has no authentication
        SPF algorithm executed 2 times
        Area ranges are
        Number of LSA 43. Checksum Sum 0x16aecb
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
    Area 2
        Number of interfaces in this area is 2
        Area has no authentication
        SPF algorithm executed 3 times
        Area ranges are
        Number of LSA 42. Checksum Sum 0x156822
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

BGP

Border Gateway Protocol version 4 (RFC 4271) is a path-vector protocol that exchanges prefixes between autonomous systems (AS). A session between routers in different AS is **eBGP** (administrative distance 20); within one AS it is iBGP (200). BGP runs over TCP port 179 between explicitly configured neighbors.

1 Design

BGP is implemented for the Head Office, as required by the brief. Core-R1 forms the Core AS and peers with each of the four Head Office routers, each in its own private AS. Using a different AS per router lets Core-R1 pass one Head Office router's prefixes to the others: if all four shared one AS, BGP loop prevention would reject those routes. OSPF remains the main IGP for the whole network.

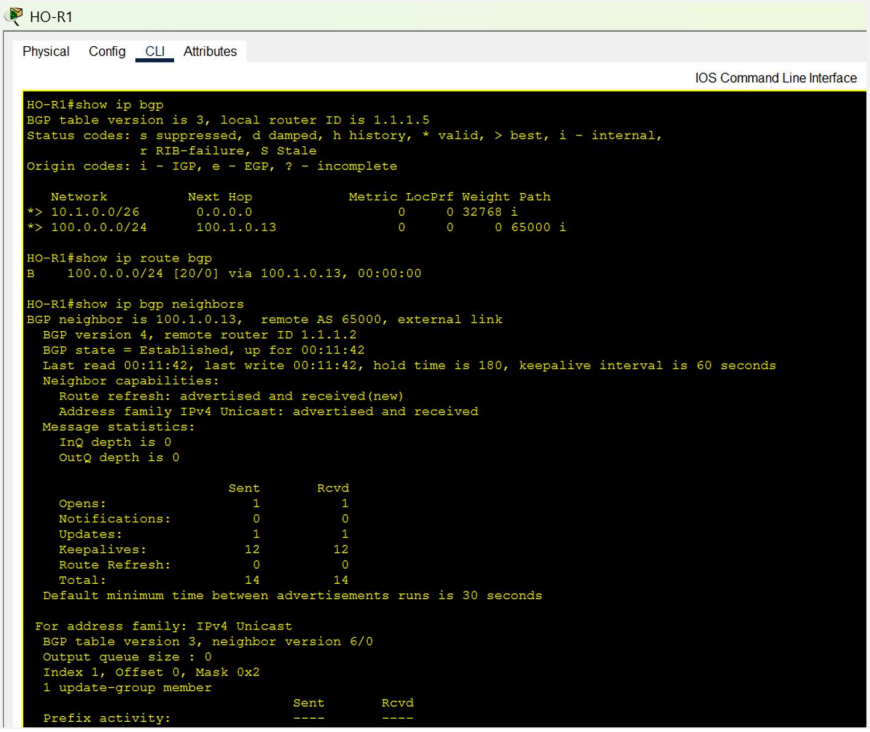
Router	AS number	Neighbor IP	Neighbor AS	Networks advertised
CORE-R1	65001	100.1.0.5, 100.1.0.9, 100.1.0.13, 100.1.0.17	65101 - 65104	None (re-advertises what it learns)
HO-R1	65101	100.1.0.5 (CORE-R1)	65001	10.1.0.0/28, 10.1.0.16/28
HO-R2	65102	100.1.0.9 (CORE-R1)	65001	10.1.0.32/28, 10.1.0.48/28
HO-R3	65103	100.1.0.13 (CORE-R1)	65001	10.1.0.64/28, 10.1.0.80/28
HO-R4	65104	100.1.0.17 (CORE-R1)	65001	10.1.0.96/28, 10.1.0.112/28

2 Interaction with OSPF

Head Office prefixes are known by both OSPF (110) and eBGP (20) on Core-R1 and the Head Office routers, so those routers install the BGP route (marked B). The next hop is the same as the OSPF path, so forwarding is unchanged. All other routers (Main, Branch, School, College) learn Head Office prefixes through OSPF only. No redistribution between OSPF and BGP is configured, because it is not needed.

3 Verification commands

show ip bgp
show ip bgp neighbors
show ip route bgp



DHCP

Dynamic Host Configuration Protocol (RFC 2131) gives clients an IP address, mask, default gateway and DNS server automatically using the four-step DORA exchange: Discover, Offer, Request, Acknowledge. Here every site router is the DHCP server for its own LANs, so clients do not depend on the WAN to obtain an address.

1 Pools

Pool	Network	Default gateway	DNS	Excluded addresses	Lease
HO-R1-LAN-A	10.1.0.0/28	10.1.0.1	100.0.0.2	10.1.0.1-10.1.0.2, 10.1.0.11, 10.1.0.12	7 days
HO-R1-LAN-B	10.1.0.16/28	10.1.0.17	100.0.0.2	10.1.0.17-10.1.0.18, 10.1.0.23	7 days
HO-R2-LAN-A	10.1.0.32/28	10.1.0.33	100.0.0.2	10.1.0.33-10.1.0.34, 10.1.0.39, 10.1.0.44	7 days
HO-R2-LAN-B	10.1.0.48/28	10.1.0.49	100.0.0.2	10.1.0.49-10.1.0.50, 10.1.0.55	7 days
HO-R3-LAN-A	10.1.0.64/28	10.1.0.65	100.0.0.2	10.1.0.65-10.1.0.66, 10.1.0.71, 10.1.0.76	7 days
HO-R3-LAN-B	10.1.0.80/28	10.1.0.81	100.0.0.2	10.1.0.81-10.1.0.82, 10.1.0.87	7 days
HO-R4-LAN-A	10.1.0.96/28	10.1.0.97	100.0.0.2	10.1.0.97-10.1.0.98, 10.1.0.103, 10.1.0.108	7 days
HO-R4-LAN-B	10.1.0.112/28	10.1.0.113	100.0.0.2	10.1.0.113-10.1.0.114, 10.1.0.119	7 days
BO-R1-LAN-A	10.2.0.0/28	10.2.0.1	100.0.0.2	10.2.0.1-10.2.0.2, 10.2.0.11, 10.2.0.12	7 days
BO-R1-LAN-B	10.2.0.16/28	10.2.0.17	100.0.0.2	10.2.0.17-10.2.0.18, 10.2.0.27	7 days
BO-R2-LAN-A	10.2.0.32/28	10.2.0.33	100.0.0.2	10.2.0.33-10.2.0.34, 10.2.0.43, 10.2.0.44	7 days
BO-R2-LAN-B	10.2.0.48/28	10.2.0.49	100.0.0.2	10.2.0.49-10.2.0.50, 10.2.0.59	7 days
BO-R3-LAN-A	10.2.0.64/28	10.2.0.65	100.0.0.2	10.2.0.65-10.2.0.66, 10.2.0.75, 10.2.0.76	7 days
BO-R3-LAN-B	10.2.0.80/28	10.2.0.81	100.0.0.2	10.2.0.81-10.2.0.82, 10.2.0.91	7 days
SHC-R1-LAN-A	10.3.0.0/28	10.3.0.1	100.0.0.2	10.3.0.1-10.3.0.2, 10.3.0.10	7 days
SHC-R1-LAN-B	10.3.0.16/28	10.3.0.17	100.0.0.2	10.3.0.17-10.3.0.18	7 days
SHC-R2-LAN-A	10.3.0.32/28	10.3.0.33	100.0.0.2	10.3.0.33-10.3.0.34, 10.3.0.42	7 days
SHC-R2-LAN-B	10.3.0.48/28	10.3.0.49	100.0.0.2	10.3.0.49-10.3.0.50	7 days
SHC-R3-LAN-A	10.3.0.64/28	10.3.0.65	100.0.0.2	10.3.0.65-10.3.0.66, 10.3.0.74	7 days
SHC-R3-LAN-B	10.3.0.80/28	10.3.0.81	100.0.0.2	10.3.0.81-10.3.0.82	7 days
COL-R1-LAN-A	10.4.0.0/28	10.4.0.1	100.0.0.2	10.4.0.1-10.4.0.2, 10.4.0.8	7 days
COL-R1-LAN-B	10.4.0.16/28	10.4.0.17	100.0.0.2	10.4.0.17-10.4.0.18	7 days
COL-R2-LAN-A	10.4.0.32/28	10.4.0.33	100.0.0.2	10.4.0.33-10.4.0.34, 10.4.0.40	7 days
COL-R2-LAN-B	10.4.0.48/28	10.4.0.49	100.0.0.2	10.4.0.49-10.4.0.50	7 days
COL-R3-LAN-A	10.4.0.64/28	10.4.0.65	100.0.0.2	10.4.0.65-10.4.0.66, 10.4.0.72	7 days
COL-R3-LAN-B	10.4.0.80/28	10.4.0.81	100.0.0.2	10.4.0.81-10.4.0.82	7 days

The pool name is the router hostname plus -LAN-A (Fa0/0 side) or -LAN-B (Fa0/1 side). Excluded addresses protect the router (.1), the switch management address (.2) and the statically addressed printers and proposed servers.

2 Lease allocation and client setup

1. On each PC or laptop open Desktop > IP Configuration and select DHCP.
2. The client broadcasts DHCPDISCOVER; the router replies with an offer from the pool of the interface that received the request.
3. The client accepts and receives address, mask, gateway and DNS 100.0.0.2; the lease lasts 7 days.
4. Verify on the router with show ip dhcp binding.

Because each LAN has 14 usable addresses and at most 9 devices plus the router and switch, no pool is expected to run out.

3 Verification commands

show ip dhcp pool
show ip dhcp binding
show ip dhcp conflict

HO-R1

Physical Config CLI Attributes

```
HO-R1#
HO-R1#show ip dhcp pool

Pool HO-R1-LAN-A :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 30
 Leased addresses               : 9
 Excluded addresses             : 2
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.1.0.1             10.1.0.1        - 10.1.0.30         9    / 2     / 30

Pool HO-R1-LAN-B :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 30
 Leased addresses               : 9
 Excluded addresses             : 2
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.1.0.33            10.1.0.33       - 10.1.0.62         9    / 2     / 30
```

HO-R1

Physical Config CLI Attributes

```
HO-R1#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.1.0.10        00D0.58CA.A98A          --                      Automatic
10.1.0.11        0001.96B2.7751          --                      Automatic
10.1.0.14        0001.9681.E2B8          --                      Automatic
10.1.0.13        00D0.BAE7.DB11          --                      Automatic
10.1.0.17        0060.3EEA.8124          --                      Automatic
10.1.0.15        000D.BD99.30B9          --                      Automatic
10.1.0.16        0060.7013.C868          --                      Automatic
10.1.0.18        00D0.BC63.C606          --                      Automatic
10.1.0.12        0060.3E2A.7C92          --                      Automatic
10.1.0.47        000C.855D.CC47          --                      Automatic
10.1.0.46        00E0.F9AC.3987          --                      Automatic
10.1.0.48        000A.416B.6DCD          --                      Automatic
10.1.0.42        00D0.BAE8.479C          --                      Automatic
10.1.0.44        0009.7C90.3461          --                      Automatic
10.1.0.45        00E0.8F6E.2C31          --                      Automatic
10.1.0.49        0060.2F67.E184          --                      Automatic
10.1.0.50        0090.0C38.D756          --                      Automatic
10.1.0.43        0006.2AC9.D64B          --                      Automatic
HO-R1#show ip dhcp conflict
IP address       Detection method   Detection time          VRF
HO-R1#
```

NAT

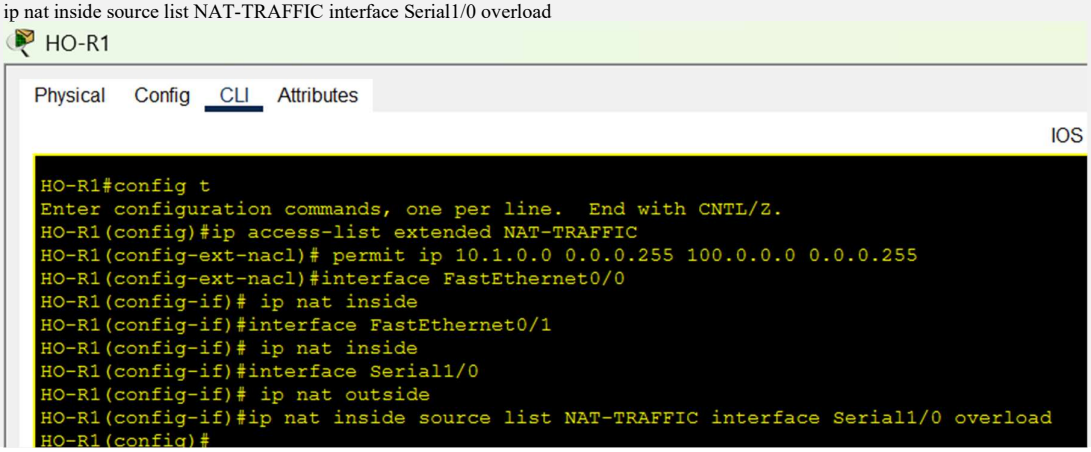
Network Address Translation (RFC 3022) rewrites IP addresses as packets cross a router. Terms: **inside local** (real host address), **inside global** (address seen by the outside network), **outside local/global** (destination as seen inside/outside). Types: static (one-to-one), dynamic (inside addresses mapped to a pool) and PAT/overload (many-to-one using port numbers).

1 Implementation at Head Office

NAT is configured on HO-R1, HO-R2, HO-R3 and HO-R4. Head Office hosts reach the central servers (100.0.0.0/24) with their source address translated to the router's own serial address. An extended ACL (NAT-TRAFFIC) selects only traffic going to the server LAN, so traffic between sites is not translated.

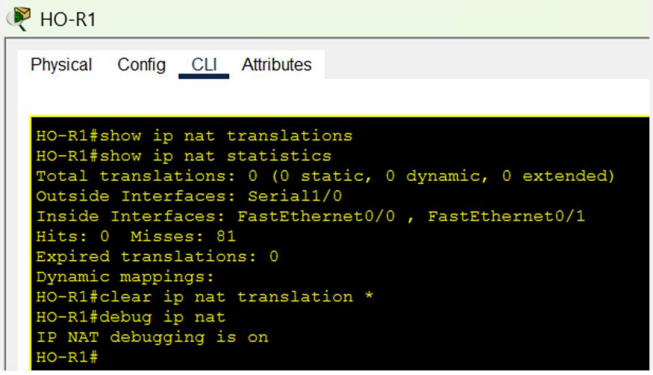
Router	Inside interfaces	Outside interface	Outside (global) IP	NAT ACL
HO-R1	F0/0, F0/1	S1/0	100.1.0.6	NAT-TRAFFIC
HO-R2	F0/0, F0/1	S1/0	100.1.0.10	NAT-TRAFFIC
HO-R3	F0/0, F0/1	S1/0	100.1.0.14	NAT-TRAFFIC
HO-R4	F0/0, F0/1	S1/0	100.1.0.18	NAT-TRAFFIC

```
ip access-list extended NAT-TRAFFIC
 permit ip 10.1.0.0 0.0.0.255 100.0.0.0 0.0.0.255
interface FastEthernet0/0
 ip nat inside
interface FastEthernet0/1
 ip nat inside
interface Serial1/0
 ip nat outside
ip nat inside source list NAT-TRAFFIC interface Serial1/0 overload
```



23.2 Verification

```
show ip nat translations
show ip nat statistics
clear ip nat translation *
debug ip nat
```



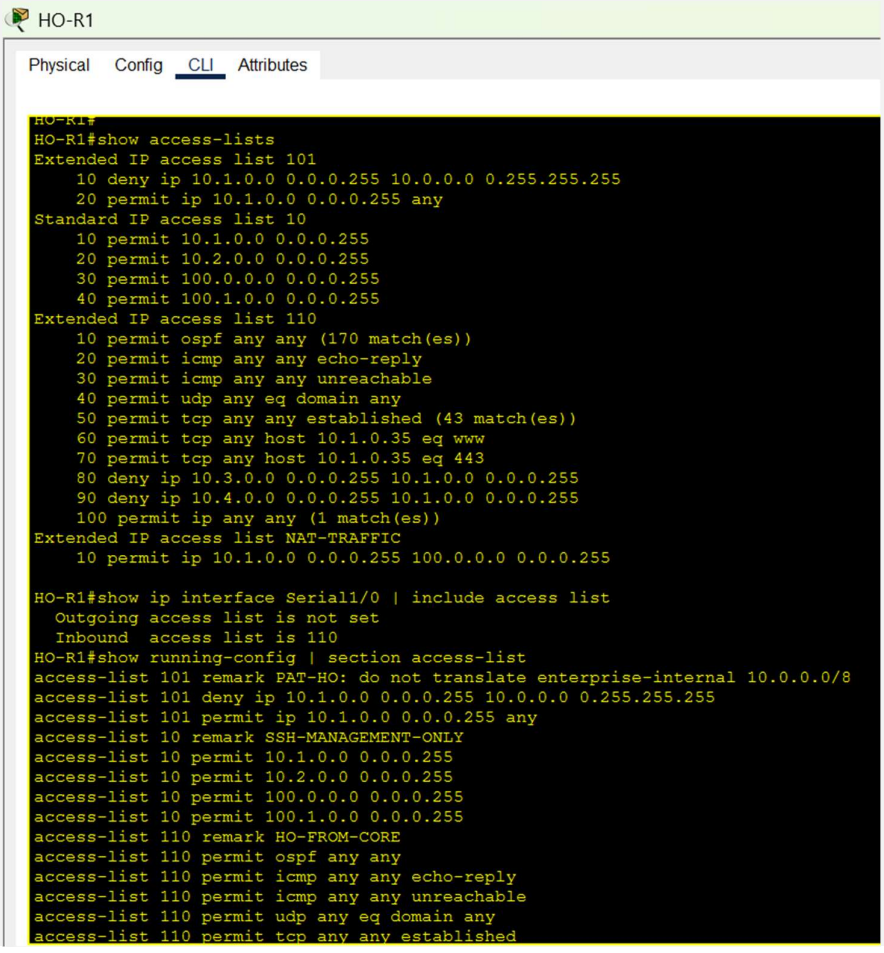
ACL

Access Control Lists filter packets by comparing them with ordered permit/deny rules; the first match wins and every list ends with an invisible **implicit deny**. A **standard** ACL (numbers 1-99) matches the source address only; an **extended** ACL matches protocol, source, destination and ports. The policy below reflects trust levels: the school is treated as an untrusted segment and may not reach internal Head Office, Branch or College LANs, and the College may not reach Head Office.

ACL	Type	Routers	Applied to	Rule	Purpose
10	Standard	HO-R1..4, BO-R1..3, COL-R1..3	VTY lines, in	permit own site /24 (e.g. 10.1.0.0 0.0.0.255); permit 100.0.0.0 0.0.0.255	SSH only from own site and Main server network
HO-IN	Extended	HO-R1..4	S1/0, in	deny ip 10.3.0.0/24 to 10.1.0.0/24; deny ip 10.4.0.0/24 to 10.1.0.0/24; permit ip any any	Block School and College from Head Office LANs
NAT-TRAFFIC	Extended	HO-R1..4	Referenced by NAT (not on an interface)	permit ip 10.1.0.0/24 to 100.0.0.0/24	Select traffic to be translated
BO-IN	Extended	BO-R1..3	S1/0, in	deny ip 10.3.0.0/24 to 10.2.0.0/24; permit ip any any	Block School from Branch LANs
COL-IN	Extended	COL-R1..3	S1/0, in	deny ip 10.3.0.0/24 to 10.4.0.0/24; permit ip any any	Block School from College LANs

Placement: each site router has one WAN link (S1/0), so filtering inbound on that interface protects every LAN behind the router. BGP, OSPF and SSH management traffic is not affected because it comes from 100.x.x.x addresses, which the lists permit.

show access-lists
show ip interface Serial1/0 | include access list
show running-config | section access-list



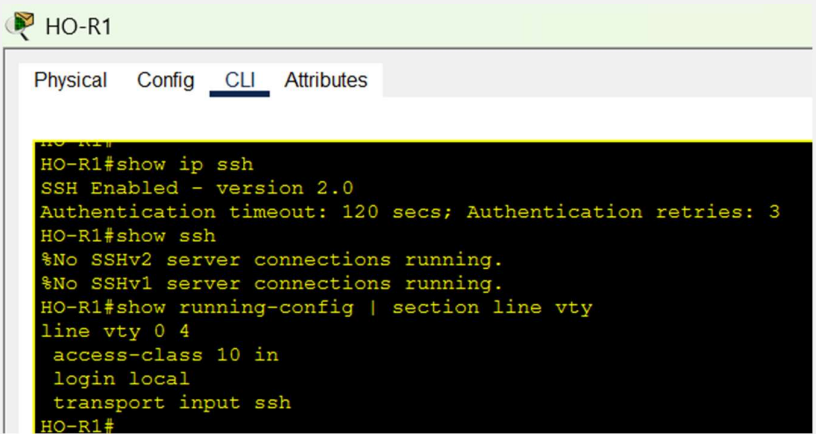
SSH

Secure Shell encrypts remote management sessions (RFC 4253) and replaces Telnet, which sends passwords in clear text. SSH needs a hostname, a domain name and an RSA key pair; version 2 is enforced. All 17 routers and 27 switches are configured the same way.

Element	Configuration
Hostname	hostname <device> (e.g. HO-R1)
Domain name	ip domain-name project.local
Username	username admin privilege 15 secret Ccna@Admin1
RSA key	crypto key generate rsa general-keys modulus 1024
Version	ip ssh version 2 (timeout 60 s, 3 retries)
VTY lines	line vty 0 4 (switches 0 15): transport input ssh, login local, exec-timeout 5 0
Restriction	access-class 10 in on Head Office, Branch and College routers

Test from a PC: open Command Prompt and run ssh -l admin 10.1.0.1 (HO-R1). Expected: password prompt, then the router prompt. Telnet to the same address should be refused.

```
show ip ssh
show ssh
show running-config | section line vty
```

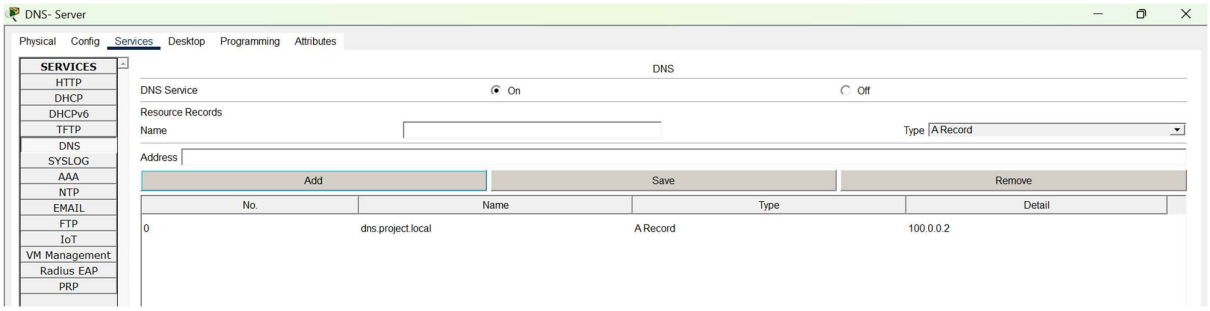


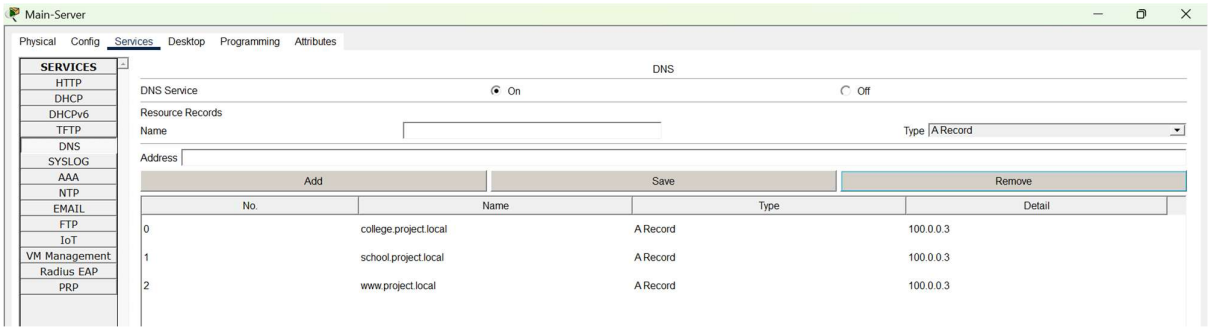
DNS

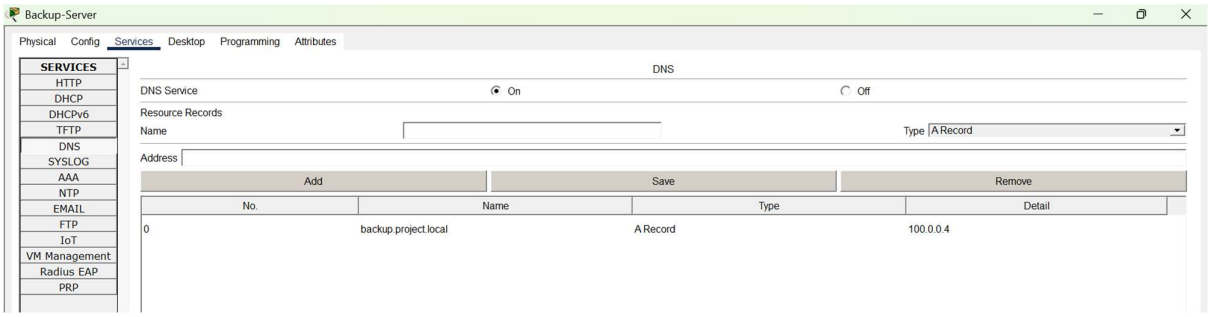
The Domain Name System (RFC 1034/1035) maps names to IP addresses so users type www.project.local instead of 100.0.0.3. One DNS server, Main Server on the Main LAN, serves all sites; DHCP hands its address to every client.

Parameter	Value
Server	Server0
Server IP	100.0.0.2 / 255.255.255.0
Default gateway	100.0.0.1 (Main - R Fa0/0)
DNS domain	project.local
Client DNS setting	100.0.0.2 (issued by DHCP; static on servers)

Name	Type	Address	Maps to
dns.project.local	A	100.0.0.2	DNS server
www.project.local	A	100.0.0.3	Web server
school.project.local	A	100.0.0.3	Web server
college.project.local	A	100.0.0.3	Web server
backup.project.local	A	100.0.0.4	Backup/FTP server



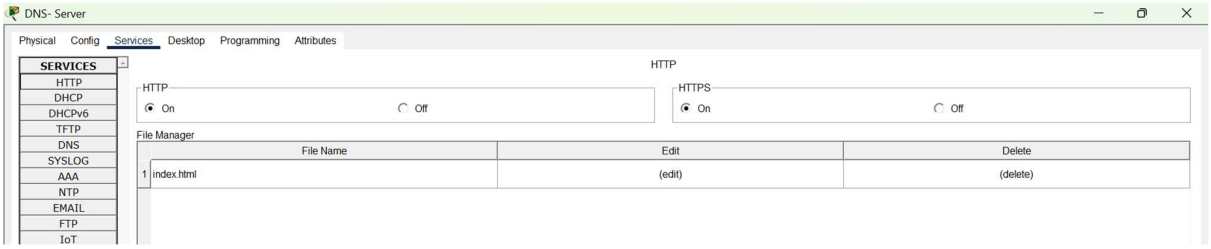




HTTP

HyperText Transfer Protocol (RFC 9110) carries web pages over TCP port 80 in clear text. The browser sends a GET request; the server answers with a status code (200 OK) and the page.

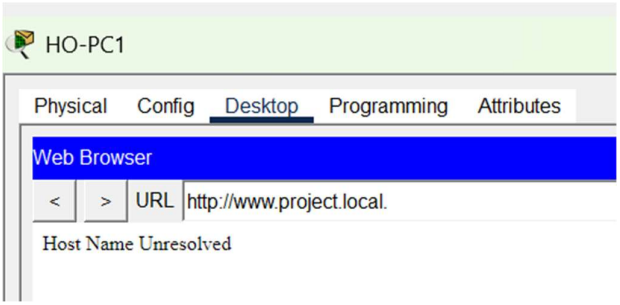
1. DNS-Server > Services > HTTP: set HTTP to On



2. Edit index.html if a custom page is wanted (for example "Integrated Enterprise Network - Main Server").



3. On a client: Desktop > Web Browser > enter http://www.project.local.



Network Testing

1 Verification commands

Area	Commands
Router basics	show ip interface brief; show ip route; show ip ospf neighbor; show ip protocols; show running-config
BGP	show ip bgp; show ip bgp summary
NAT	show ip nat translations; show ip nat statistics
DHCP	show ip dhcp binding; show ip dhcp pool
ACL	show access-lists
SSH	show ip ssh
DNS / HTTP / HTTPS	Client testing: on a PC run ipconfig /all (check DNS 100.0.0.2), nslookup www.project.local (expect 100.0.0.3), then Web Browser: http://www.project.local and https://www.project.local. If the IP works but the name does not, check the DNS server records and service switch.

Advantages

- Hierarchical design that is easy to understand and to extend by adding site routers under a Core router.
- Multi-area OSPF keeps routing tables and updates local to each Core domain.
- VLSM addressing wastes no space: /30 for point-to-point links and /28 for LANs.
- Central DNS and web services reduce administration; router-based DHCP keeps sites independent.
- Layered security: ACLs, SSH, hardened logins and Layer 2 protections.
- Every configuration value is traceable to one addressing plan.

Limitations

- The tree topology has no redundancy: loss of one serial link isolates every site below it.
- Only three servers are cabled; the 13 per-site servers cannot provide services until connected.
- No VLANs exist, so all devices in a LAN share one broadcast domain and there is no segmentation between departments.
- NAT and BGP are used at Head Office mainly to demonstrate the technologies; OSPF alone would route the network.
- Serial links run at 2 Mbit/s in the simulation and Packet Tracer does not model real-world load, latency or hardware limits.
- Lab passwords and RSA 1024-bit keys are acceptable for the project but not for production.
- Test results have not been captured; all Actual Result fields remain to be verified.

Future Scope

- Add redundant links between Core routers and dual-home site routers (then STP/OSPF failover can be demonstrated).
- Introduce VLANs, inter-VLAN routing and port security.
- Add a firewall or zone-based policy firewall, AAA with RADIUS/TACACS+, and syslog/NTP.
- Add IPv6, HSRP/VRRP gateway redundancy and QoS.
- Cable and configure the per-site servers and the ATM site found in the project file.
- Replace the lab simulation with GNS3/EVE-NG or physical devices.

Conclusion

This document records the Integrated Enterprise Network exactly as cabled in the author's Packet Tracer project: 17 routers, 27 switches, 156 PCs, 28 laptops, 14 printers and 16 servers. Differences between the brief, the screenshot and the project file were identified and listed, including a router count of 17 rather than 16 and an extra ATM site present only in the project file.

A single addressing plan drives every table, routing example and CLI block, so the documentation is internally consistent. The design provides multi-area OSPF, eBGP at Head Office, DHCP, DNS, HTTP/HTTPS, NAT/PAT, ACLs, SSH and STP with defined verification commands and a 14-test plan. The next step is to apply the configuration in Packet Tracer, run the tests and record the actual results.

References

1. Cisco Systems. Cisco Packet Tracer 9.0 documentation and Cisco Networking Academy course material.
2. ChatGPT
3. Claude AI
4. GitHub